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Syllabus

Course Contents / Syllabus		
Module 1	**INTRODUCTION TO MACHINE LEARNING**	**10 hours**
Learning, Types of Learning, well defined learning problems, designing a Learning System, History of ML, Introduction of Machine Learning Approaches, Features and their types, handing missing data, Dealing with categorical features, Feature Scaling, Feature selection, outlier detection and removal. Introduction to Model Building, Sensitivity Analysis, Underfitting and Overfitting, Bias and Variance, Activation Functions, Data Science Vs Machine Learning.		
Module 2	**CLASSIFICATION**	**18 hours**
Classification: Classification problem, Binary and Multiclass Classification, Decision Boundary **Measuring Performance of Classification Model:** Accuracy, Precision, Recall, F1-Score, ROC-AUC curve, Confusion matrix. **Decision Trees:** Decision Tree Classification Algorithm, Attribute Selection Measures, Variants of Decision Trees. **Large Margin Classifier:** Support Vector Machines, **Instance Based Learning:** K Nearest Neighbor Algorithm.		
Module 3	**REGRESSION**	**12 hours**
Logistic regression model for binary classification, Cost function of logistic regression model, Regularized logistic regression, One Vs. All strategy for multiclass classification using logistic regression. **Introduction to Regression:** Regression Problem, Linear regression model (Simple, Multiple and Polynomial), **Errors Measures:** Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), **Cost function:** Formula, intuition and visualization, Bias and Variance, Overfitting		

Syllabus

and Underfitting, **Gradient Descent:** Algorithm, intuition, learning rate and Gradient Descent for linear regression.		
Module 4	**PROBABILISTIC & CLUSTERING TECHNIQUES**	**12 hours**
Probabilistic Methods for Classification: Bayesian Learning, Naive Bayes Classifier, Bayesian belief networks. **Tree Ensembles:** Using multiple decision trees, Bagging Vs. Boosting, Sampling with replacement, Random Forest Algorithm, Adda boost, Gradient Boosting, Xgboost Algorithm. **Unsupervised Learning &Clustering:** The problem of Feed, Distance measures **Partition Based Methods:** K-Means Clustering (Intuition, Algorithm, Optimization objective, Initializing K-Means, Choosing number of clusters) K-Medoids and K-Mode clustering algorithms. **Hierarchical Clustering** (Single Linkage, Complete Linkage, AGNES (Agglomerative Nesting), DIANA (Divisive Analysis). Reinforcement Learning		
Module 5	**ARTIFICIAL NEURAL NETWORKS**	**8 hours**
Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Feed Forward Networks, Back Propagation Networks, Back-propagation through time (BTT), Different types of RNNs. Recurrent Neural Network Model, Notation, hyper-parameter tuning, Image classification, Text classification, Image classification. Understanding and visualizing a CNN. Case Study: Health Care, E Commerce, Smart Cities.		
Total Lecture Hours		**60 hours**

Branch wise Applications

- Artificial Intelligence and Robotics
- Cybersecurity
- Natural Language Processing (NLP)
- Computer Vision
- Software Engineering
- Data Mining and Big Data Analytics
- Bioinformatics and Healthcare
- Recommender Systems
- Cloud and Edge Computing
- Internet of Things (IoT)

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Course Objective

- To introduce students to the basic concepts of Machine Learning.
- To develop skills of implementing machine learning for solving practical problems.
- To gain experience of doing independent study and research related to Machine Learning

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Course Outcomes

Course outcome: After completion of this course students will be able to:		
CO1	Understanding utilization and implementation proper machine learning algorithm.	K2
CO2	Understand the basic supervised machine learning algorithms.	K2
CO3	Understand the difference between supervise and unsupervised learning.	K2
CO4	Understand algorithmic topics of machine learning and mathematically deep enough to introduce the required theory.	K2
CO5	Apply an appreciation for what is involved in learning from data.	K3

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Program Outcomes

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

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Program Outcomes

Engineering Graduates will be able to:

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

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Program Outcomes

Engineering Graduates will be able to:

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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COs and POs Mapping

Machine Learning Techniques (BCSML0501A)									Year of Study: 2025-26			
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
BCSML0501A .1	3	2	2	1	2	2	-	-	-	1	-	-
BCSML0501A.2	3	2	2	3	2	2	1	-	2	1	1	2
BCSML0501A.3	2	2	2	2	2	2	2	1	1	-	1	3
BCSML0501A.4	3	3	1	3	1	1	2	-	2	1	1	2
BCSML0501A.5	3	2	1	2	1	2	1	1	2	1	1	1
Average	2.8	2.2	1.6	2.2	1.6	1.8	1.2	0.4	1.4	0.8	0.8	1.6

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Program Specific Outcomes (PSOs)

On successful completion of graduation degree the Computer Science & Engineering graduates will be able to:

PSO1:
Identify, analyze real world problems and design their ethical solutions using artificial intelligence, robotics, virtual/augmented reality, data analytics, block chain technology, and cloud computing.

PSO2:
Design and develop the hardware sensor devices and related interfacing software systems for solving complex engineering problems.

PSO3:
Understand inter-disciplinary computing techniques and to apply them in the design of advanced computing.

PSO4:
Conduct investigation of complex problems with the help of technical, managerial, leadership qualities, and modern engineering tools provided by industry-sponsored laboratories

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COs and PSOs Mapping

CO	Program Specific Outcomes			
	PSO1	PSO2	PSO3	PSO4
BCSML0501A .1	**3**	**2**	**3**	**3**
BCSML0501A.2	3	2	2	3
BCSML0501A.3	3	2	3	3
BCSML0501A.4	2	1	1	2
BCSML0501A.5	2	2	1	2
Average	2.6	1.8	2	2.6

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Program Educational Objectives (PEOs)

The graduates will be:

PEO1: To have an excellent scientific and engineering breadth so as to comprehend, analyze, design and provide sustainable solutions for real-life problems using state-of-the-art technologies.

PEO2: To have a successful career in industries, to pursue higher studies or to support entrepreneurial endeavors and to face the global challenges.

PEO3: To have an effective communication skill, professional attitude, ethical values and a desire to learn specific knowledge in emerging trends, technologies for research, innovation and product development and contribution to society.

PEO4: To have life-long learning for up-skilling and re-skilling for a successful professional career as an engineer, scientist, entrepreneur or bureaucrat for the betterment of the society.

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Result Analysis

Subject Name	Section	Faculty Name	Total Students	Pass Students	% Section wise Result	Overall % Result of Subject		Highest Marks	Average Marks
Machine Learning BCSML0501	A	MS. SANA ANJUM	68	68	100	99.63		**83**	**60.81**
	B	MS. SANA ANJUM	67	66	98.51			**86**	**58.01**
	C	DR. VIVEK	68	68	100			**77**	**59.38**
	D	MR. PRADEEP KUMAR	65	65	100			**87**	**57.04**

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Question Paper Template

PAPER ID-310687

Roll No:

B. TECH
(SEM-V) THEORY EXAMINATION 2020-21
MACHINE LEARNING TECHNIQUES

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

Qno.	Question	Marks	CO
a.	Explain the concept of machine learning.	2	1
b.	Compare ANN and Bayesian networks.	2	4
c.	What is the difference between linear and logistic regression?	2	2
d.	Discuss support vectors in SVM?	2	2
e.	Discuss overfitting and underfitting situation in decision tree learning.	2	3
f.	What is the task of the E-step of the EM-algorithm?	2	2
g.	Define the learning classifiers.	2	2
h.	What is the difference between machine learning and deep learning?	2	1
i.	What objective function do regression trees minimize?	2	2
j.	What is the difference between Q learning and deep Q-learning?	2	5

SECTION B

2. Attempt any *three* of the following:

Qno.	Question	Marks	CO
a.	Apply KNN for following dataset and predict class of test example (A1=3, A2=7).Assume K=3	10	3
b.	Describe the Kohonen Self-Organizing Maps and its algorithm.	10	4
c.	Explain the various learning models for reinforcement learning.	10	5
d.	Explain the role of genetic algorithm? Discuss the various phases considered in genetic algorithm.	10	5
e.	Describe BPN algorithm in ANN along with a suitable example.	10	4

A1	A2	Class
7	7	True
7	4	True
3	4	False
1	4	True
5	3	False
6	3	True

SECTION C

3. Attempt any *one* part of the following:

Qno.	Question	Marks	CO
a.	Why SVM is an example of a large margin classifier? Discuss the different kernels functions used in SVM.	10	2
b.	Explain the relevance of CBR. How CADET tool employs CBR?	10	3

4. Attempt any *one* part of the following:

Qno.	Question	Marks	CO
a.	Discuss the applications, properties, issues, and disadvantages of SVM.	10	2
b.	Explain the Confusion Matrix with respect to Machine Learning Algorithms.	10	1

PAPER ID-310687

Roll No:

5. Attempt any *one* part of the following:

a.	Illustrate the operation of the ID3 training example. Consider information gain as attribute measure.	10	3
b.	Describe Markov Decision Process in reinforcement learning.	10	5

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

6. Attempt any *one* part of the following:

Qno.	Question	Marks	CO
a.	What is instance based learning? How Locally Weighted Regression is different from Radial basis function networks?	10	3
b.	How is Bayes theorem used in machine learning? How naive Bayes algorithm is different from Bayes theorem?	10	2

7. Attempt any *one* part of the following:

Qno.	Question	Marks	CO
a.	Compare regression, classification and clustering in machine learning along with suitable real life applications?	10	1
b.	Given below is an input matrix named I, kernel matrix, calculate the Convoluted matrix C using stride =1 also apply max pooling on C.	10	4

Input Matrix I

1	0	0	1	1	0	1	
0	0	1	1	1	0	1	
1	1	1	0	1	0	1	
1	1	0	1	0	0	0	
1	0	1	0	1	1	0	
0	1	1	0	0	1	1	
0	1	1	1	0	1	1	

Kernel Matrix

1	0	0
0	1	1
1	1	0

Prerequisite and Recap

Prerequisites:

- Statistics.
- Linear Algebra.
- Calculus.
- Probability.
- Programming Languages.

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Brief Introduction about the Subject

What is Machine Learning?

"Learning is any process by which a system improves performance from experience."

- Herbert Simon

Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E.

A well-defined learning task is given by $<P, T, E>$.

https://www.youtube.com/watch?v=ukzFI9rgwfU

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Content – Unit 1

INTRODUCTION – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches, Introduction to Model Building, Sensitivity Analysis, Underfitting and Overfitting, Bias and Variance, Concept Learning Task, Find – S Algorithms, Version Space and Candidate Elimination Algorithm, Inductive Bias, Issues in Machine Learning and Data Science Vs Machine Learning.

Unit Objective

The objective of the Unit 1 is

1. To understand the basics of machine learning,

2. A clear concept of Learning and learning systems

3. Brief History of ML

4. Use of various approaches of Machine learning

5. understand the difference between Data Science and Machine learning

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Topic Objective

- Student will be able to understand
- Learning
- Types of Learning,
- Well defined learning problems,
- Designing a Learning System

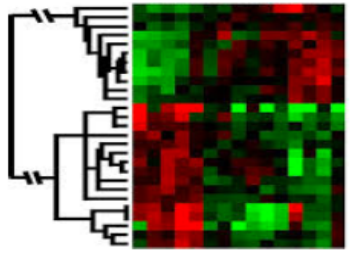
Introduction

When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



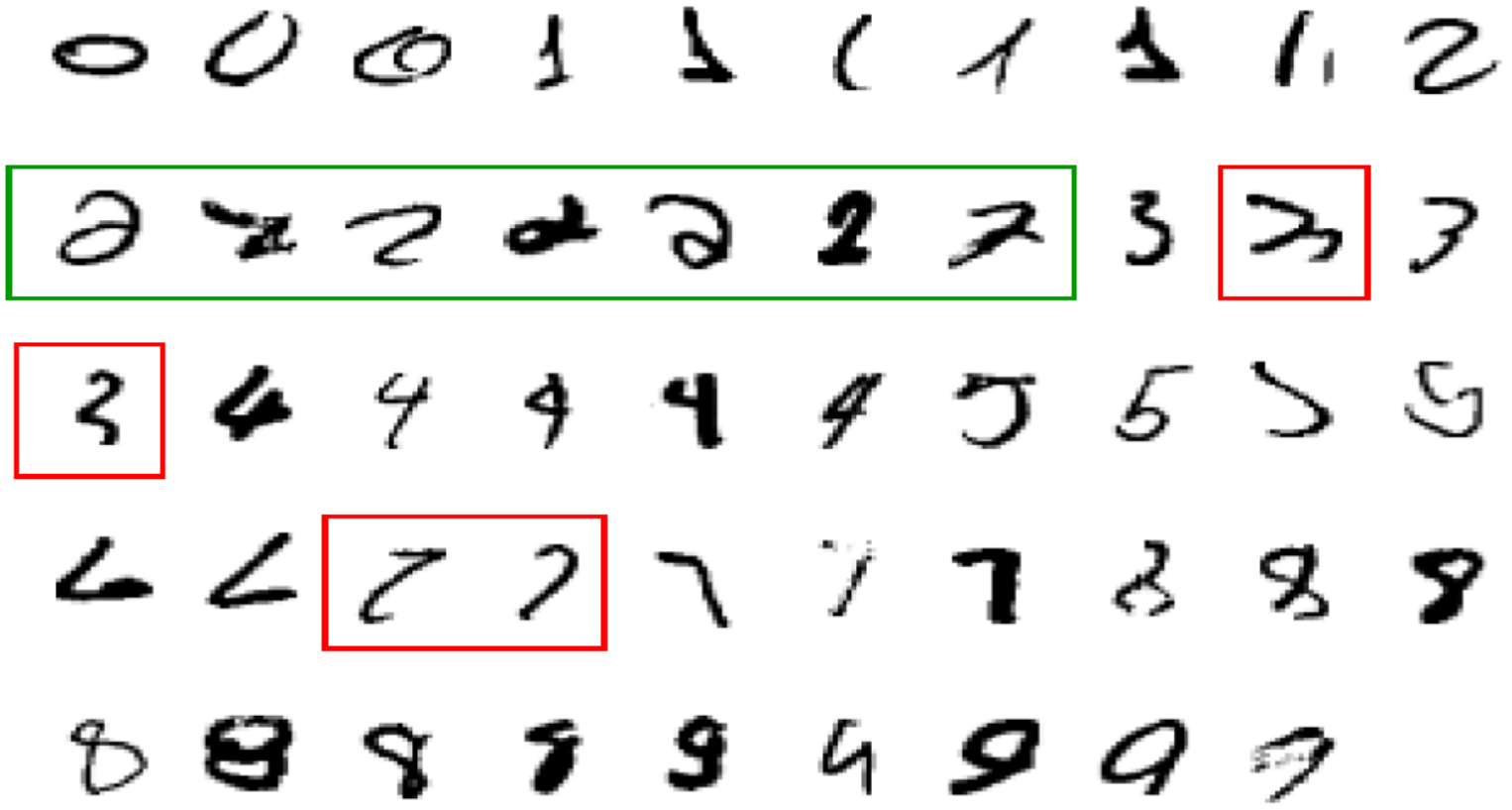
Learning isn't always useful:

- There is no need to "learn" to calculate payroll



Example

A classic example of a task that requires machine learning:
It is very hard to say what makes a 2





Example

Some more examples of tasks that are best solved by using a learning algorithm

- Recognizing patterns:
  - Facial identities or facial expressions
  - Handwritten or spoken words
  - Medical images
- Generating patterns:
  - Generating images or motion sequences
- Recognizing anomalies:
  - Unusual credit card transactions
  - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
  - Future stock prices or currency exchange rates

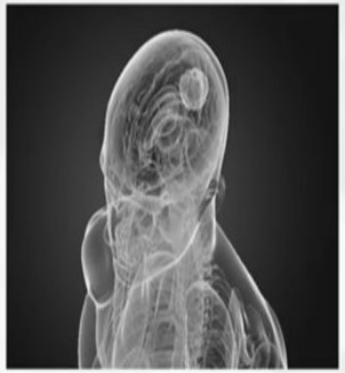


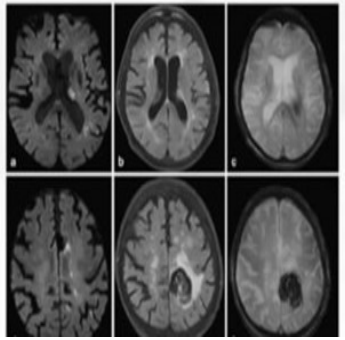
Example



GOOGLE MAPS PREDICTS WHETHER THE TRAFFIC IS CLEAR, SLOW-MOVING OR HEAVILY CONGESTED BASED ON TWO MEASURES:

- AVERAGE TIME TAKEN ON SPECIFIC DAYS AT SPECIFIC TIMES ON THAT ROUTE
- REAL TIME LOCATION DATA OF VEHICLES FROM GOOGLE MAPS APPLICATION AND SENSORS





MEDICAL TECHNOLOGY HAS INNOVATED WITH THE USE OF MACHINE LEARNING TO DIAGNOSE DISEASES

WE ARE ABLE TO ANALYZE 2D CT SCANS AND COME UP WITH 3D MODELS THAT PREDICT WHERE EXACTLY THERE ARE LESIONS IN THE BRAIN

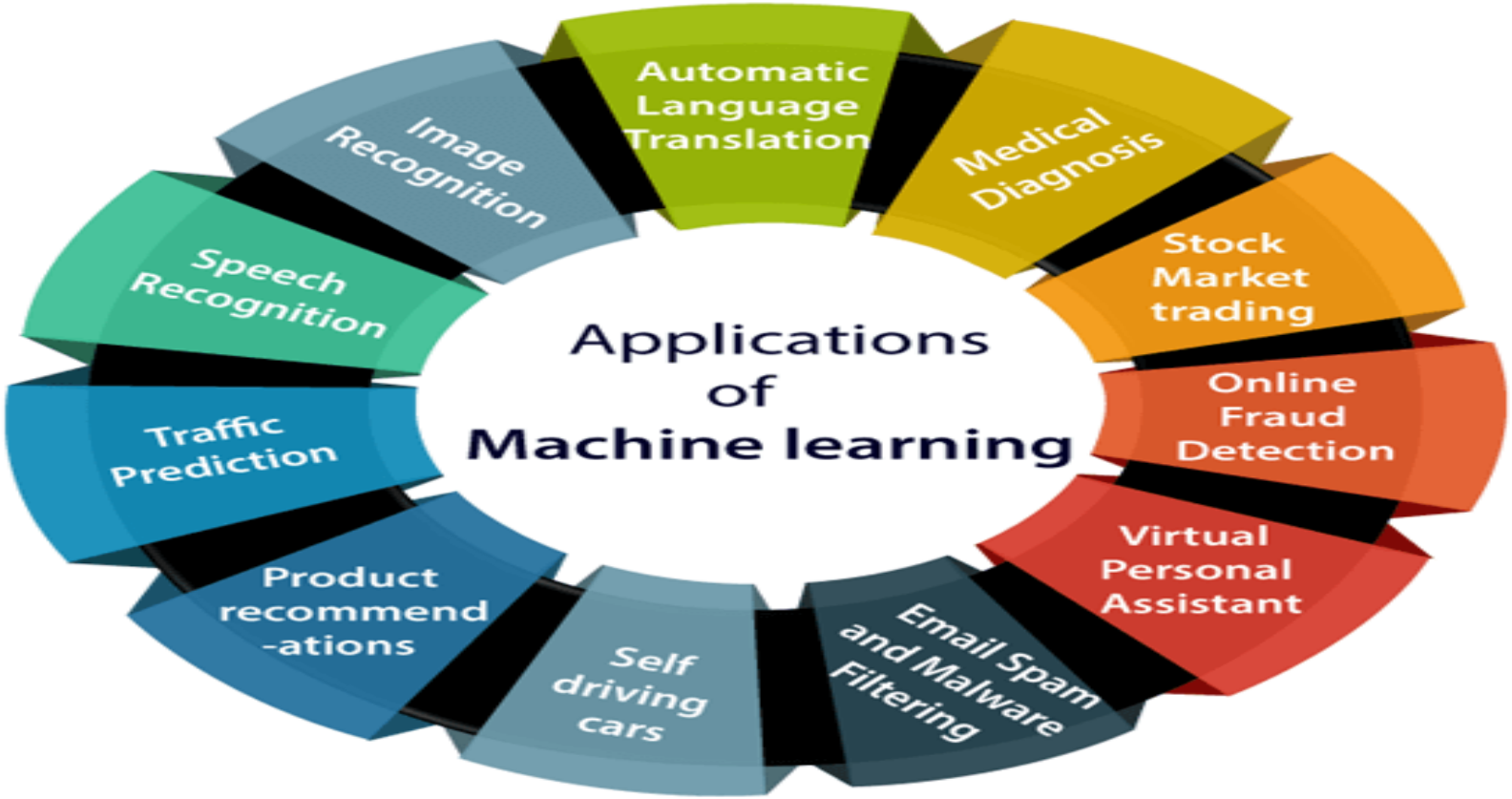
IT WORKS EQUALLY WELL FOR BRAIN TUMORS AND ISCHEMIC STROKE LESIONS

THIS CAN BE FURTHER USED IN FETAL IMAGING AND CARDIAC ANALYSIS TOO

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Applications of Machine learning

Machine learning is a buzzword for today's technology, and it is growing very rapidly day by day.





QUIZ

1. When you hear ‘Machine Learning’, what’s the first word that comes to your mind?
2. Name an everyday example where you think ML might be used.
3. Which of the following best describes Machine Learning?

   a) Writing rules for every possible scenario
   b) Systems that learn patterns from data
   c) Random guessing
   d) Manual programming for each task

QUIZ

4. Which is **not** a type of Machine Learning?
   a) Supervised
   b) Unsupervised
   c) Reinforcement
   d) Reverse Learning

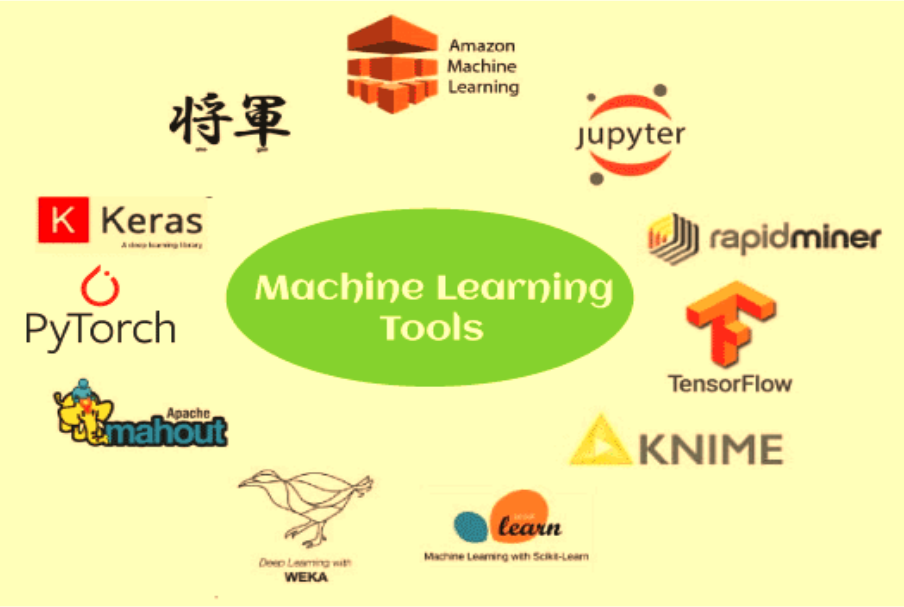
5. Match the **ML type** to its example:

Supervised →	Customer segmentation
Unsupervised →	Email spam detection
Reinforcement →	Robot learning to walk

Learning

Machine learning (ML) is a subdomain of artificial intelligence (AI) that focuses on developing systems that learn—or improve performance—based on the data they ingest.

- ***It is a subfield of Artificial Intelligence, which analyses the data, build the model, and make predictions***.



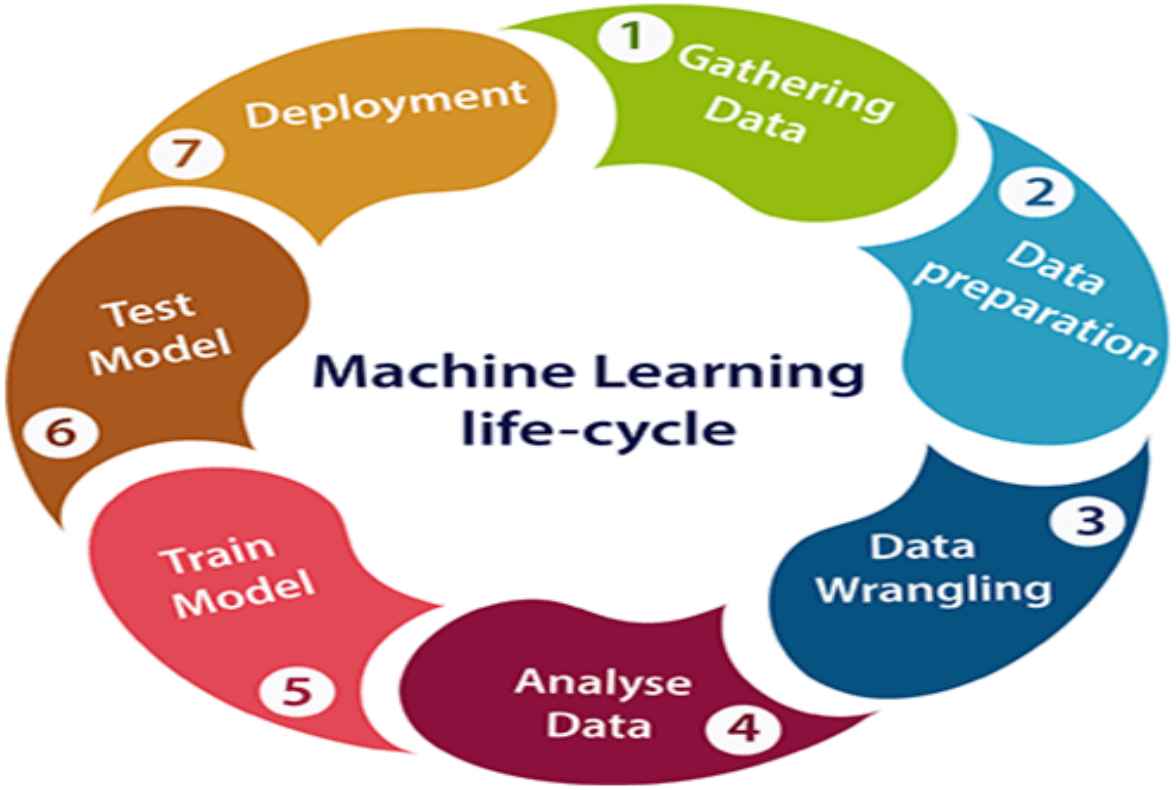


Learning

Machine learning Life-cycle

Machine learning life cycle is a cyclic process to build an efficient machine learning project.

Machine learning life cycle involves seven major steps, which are given below:





Learning

1. **Gathering Data**
2. **Data preparation**
3. **Data Wrangling** (transforming raw data into more usable form)
4. **Analyse Data**
5. **Train the model**
6. **Test the model**
7. **Deployment**



Learning

What Is Learning?

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

The ability to learn is possessed by humans, non-human animals, and some machines

Human learning starts at birth and continues until death as a consequence of ongoing interactions between people and their environment.





Types of Machine Learning

Types of Learning

- **Supervised (inductive) learning**
  — Given: training data + desired outputs (labels)
- **Unsupervised learning**
  — Given: training data (without desired outputs)
- **Semi-supervised learning**
  — Given: training data + a few desired outputs
- **Reinforcement learning**
  — Rewards from sequence of actions



Types of Learning

On basis of the nature of the learning "signal" or "feedback" available to a learning system

I. Supervised learning:

The model or algorithm is presented with example inputs and their desired outputs and then finds patterns and connections between the input and the output.

Image Classification:
Market Prediction/Regression:

II. Unsupervised learning:

No labels are given to the learning algorithm, leaving it on its own to find structure in its input.

Clustering:
High-Dimension Visualization:
Generative Models:



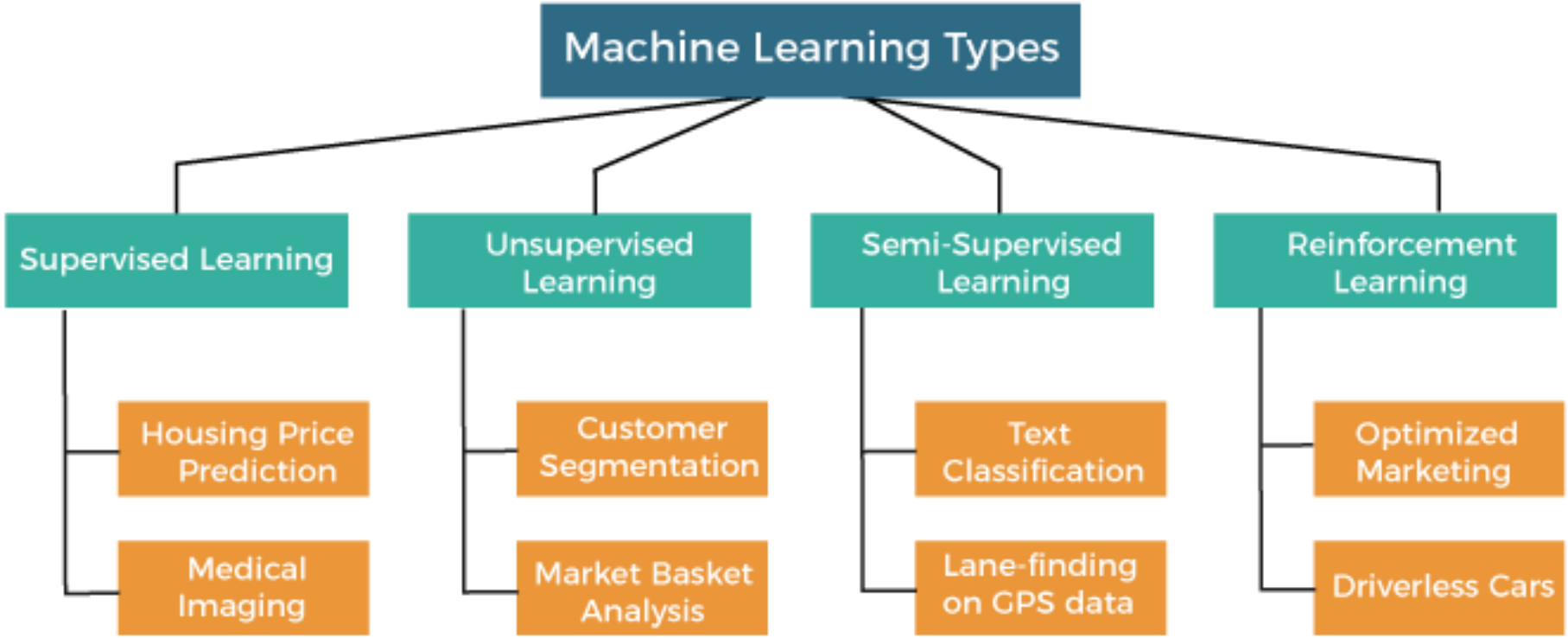
Types of Learning

On basis of the nature of the learning "signal" or "feedback" available to a learning system

III. Semi-supervised learning:
Problems where you have a large amount of input data and only some of the data is labelled, are called semi-supervised learning problems.

III. Reinforcement learning:
A computer program interacts with a dynamic environment in which it must perform a certain goal

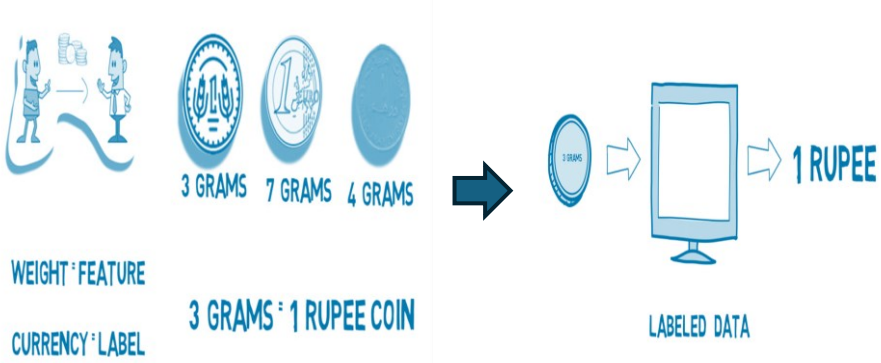



Machine Learning Types
Supervised Learning
Unsupervised Learning
Semi-Supervised Learning
Reinforcement Learning
Housing Price Prediction
Medical Imaging
Customer Segmentation
Market Basket Analysis
Text Classification
Lane-finding on GPS data
Optimized Marketing
Driverless Cars

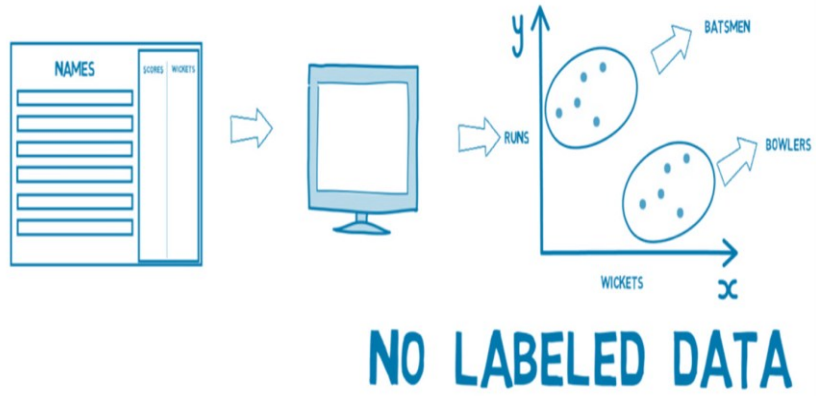


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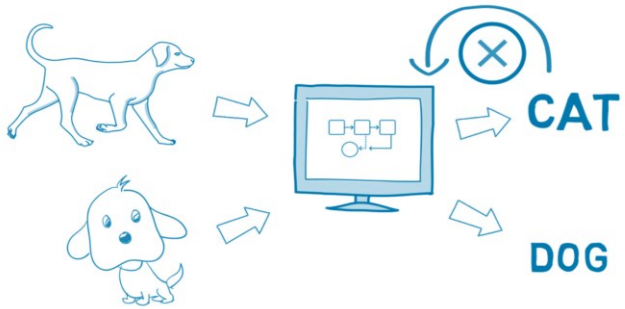
Supervised Learning



Supervised Learning



Unsupervised Learning



Reinforcement Learning

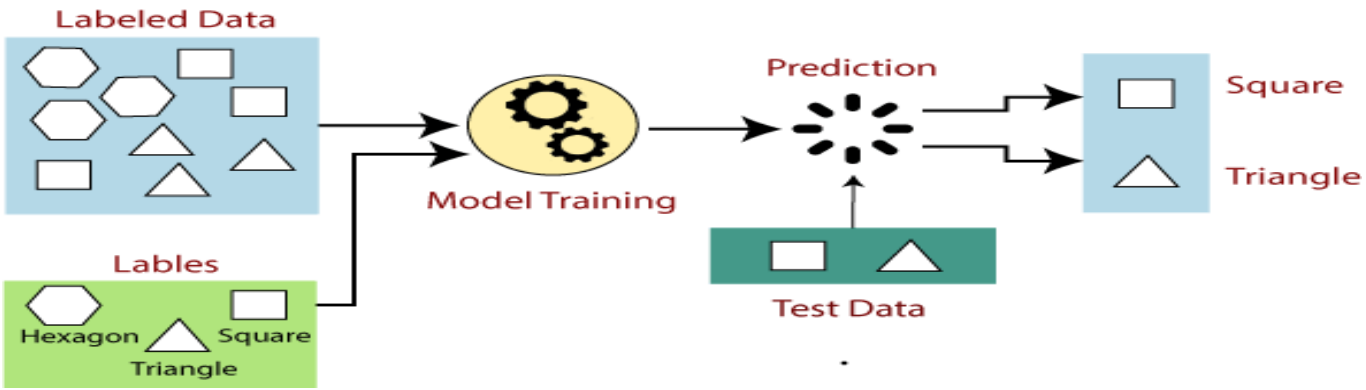


Supervised Learning



Supervised Machine Learning

Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of that data, machines predict the output.



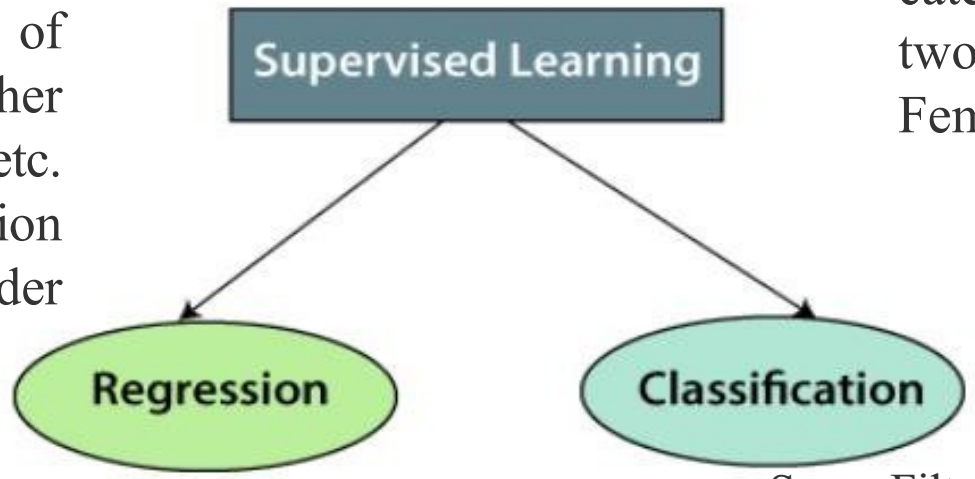


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Supervised Learning

Regression

Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc. Below are some popular Regression algorithms which come under supervised learning:



- Linear Regression
- Regression Trees
- Non-Linear Regression
- Bayesian Linear Regression
- Polynomial Regression

Classification

Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

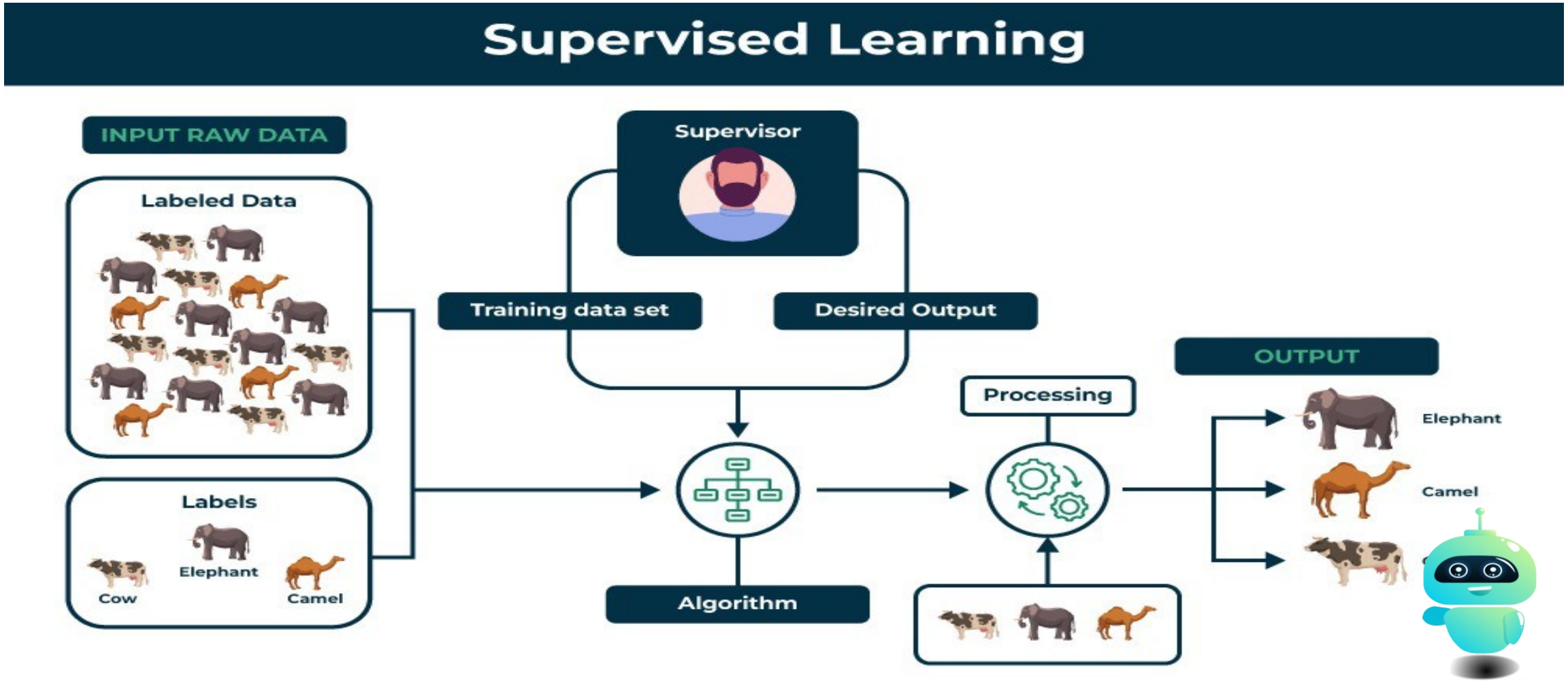
Spam Filtering,

- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines



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Supervised Learning



Advantages and Disadvantages of Supervised Learning

Advantages:

- Since supervised learning work with the labelled dataset so we can have an exact idea about the classes of objects.
- These algorithms are helpful in predicting the output on the basis of prior experience.

Disadvantages:

- These algorithms are not able to solve complex tasks.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.



Applications of Supervised Learning

Some common applications of Supervised Learning are given below:

•**Image Segmentation:**

Supervised Learning algorithms are used in image segmentation. In this process, image classification is performed on different image data with pre-defined labels.

•**Medical Diagnosis:**

Supervised algorithms are also used in the medical field for diagnosis purposes. It is done by using medical images and past labelled data with labels for disease conditions. With such a process, the machine can identify a disease for the new patients.

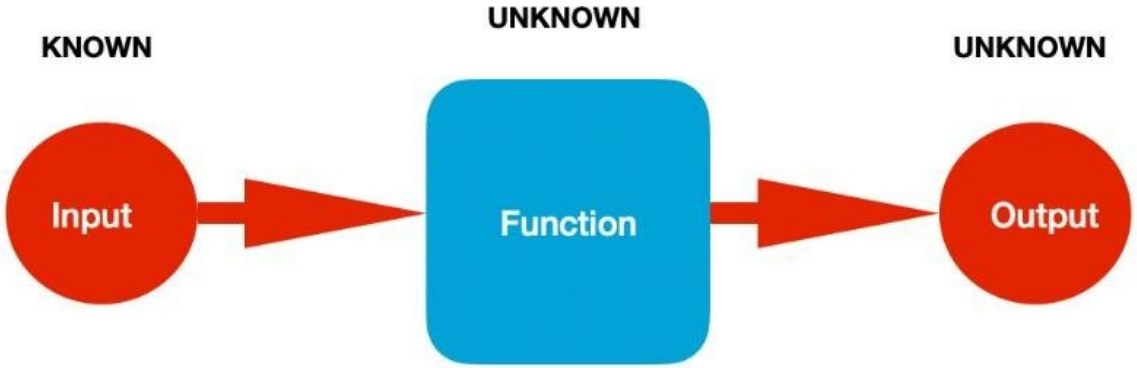
•**Fraud Detection -** Supervised Learning classification algorithms are used for identifying fraud transactions, fraud customers, etc. It is done by using historic data to identify the patterns that can lead to possible fraud.

•**Spam detection -** In spam detection & filtering, classification algorithms are used. These algorithms classify an email as spam or not spam. The spam emails are sent to the spam folder.

•**Speech Recognition -** Supervised learning algorithms are also used in speech recognition. The algorithm is trained with voice data, and various identifications can be done using the same, such as voice-activated passwords, voice commands, etc.

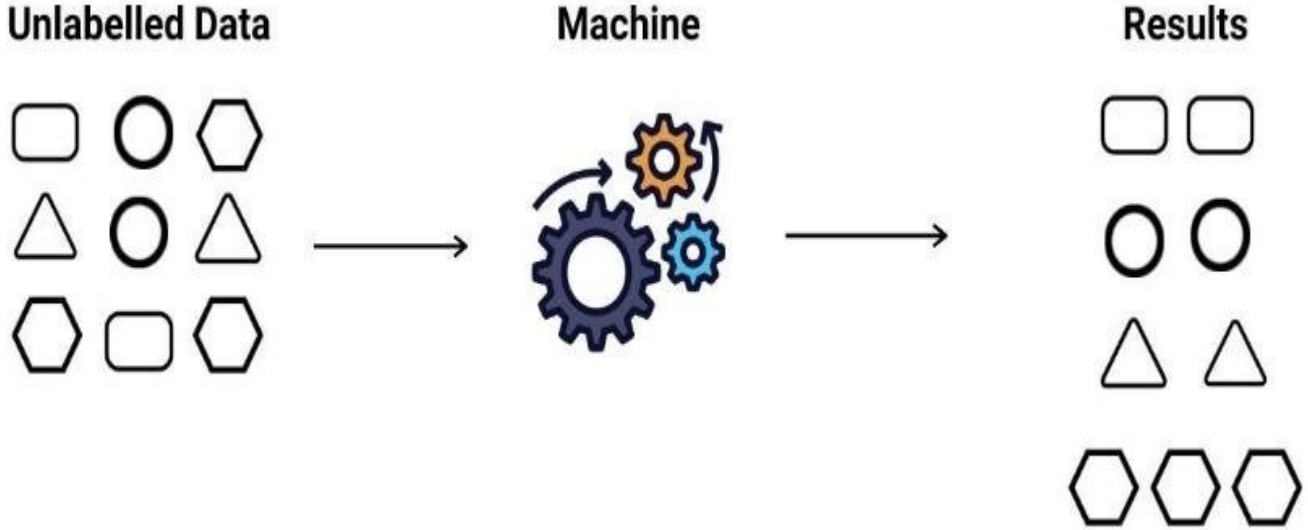
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Unsupervised Learning



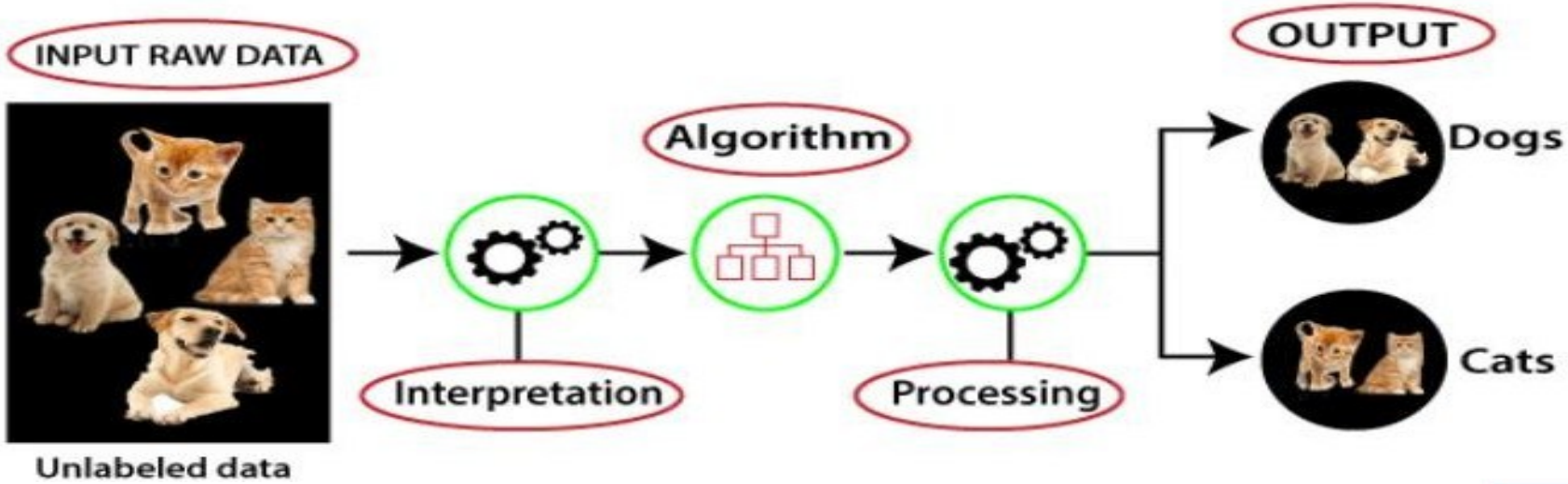
Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision

The goal of unsupervised learning is to **find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format**.



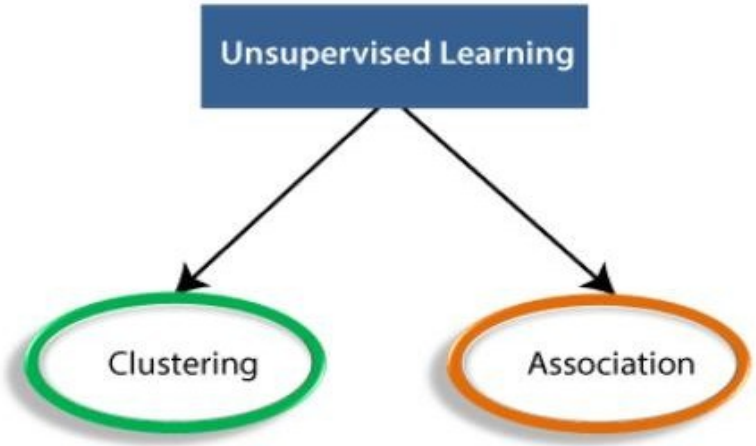
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Unsupervised Learning



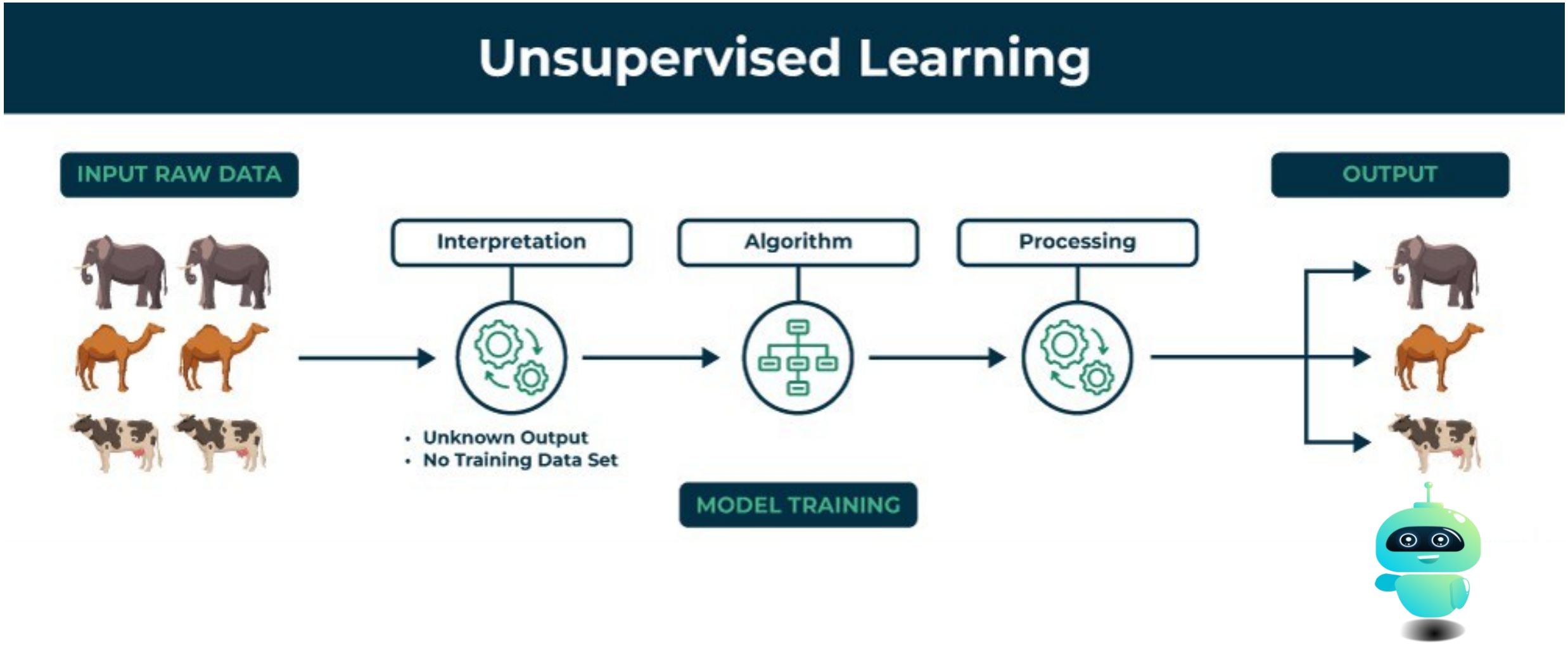
Types of Unsupervised Learning Algorithm:

The unsupervised learning algorithm can be further categorized into two types of problems:



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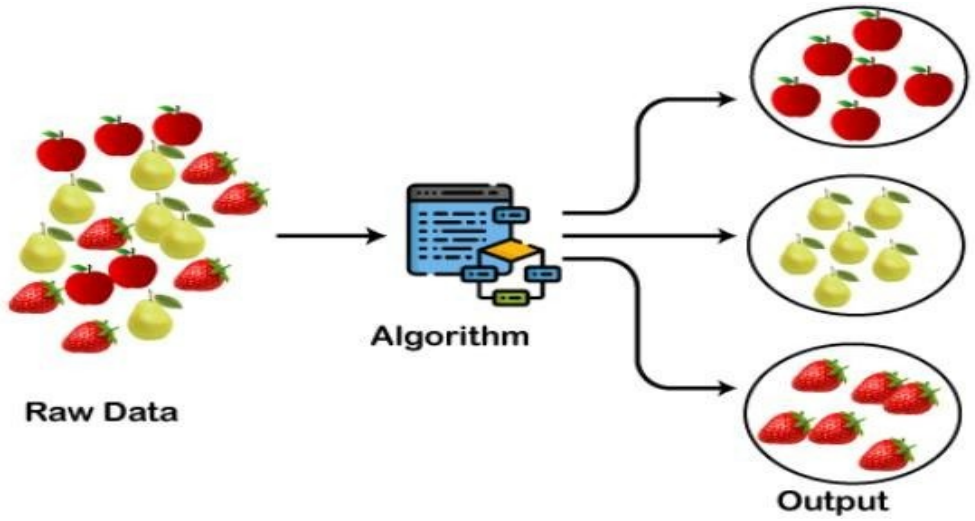
Unsupervised Learning



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Unsupervised Learning

Clustering: Clustering is a method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group. Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.



Association: An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset. Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.

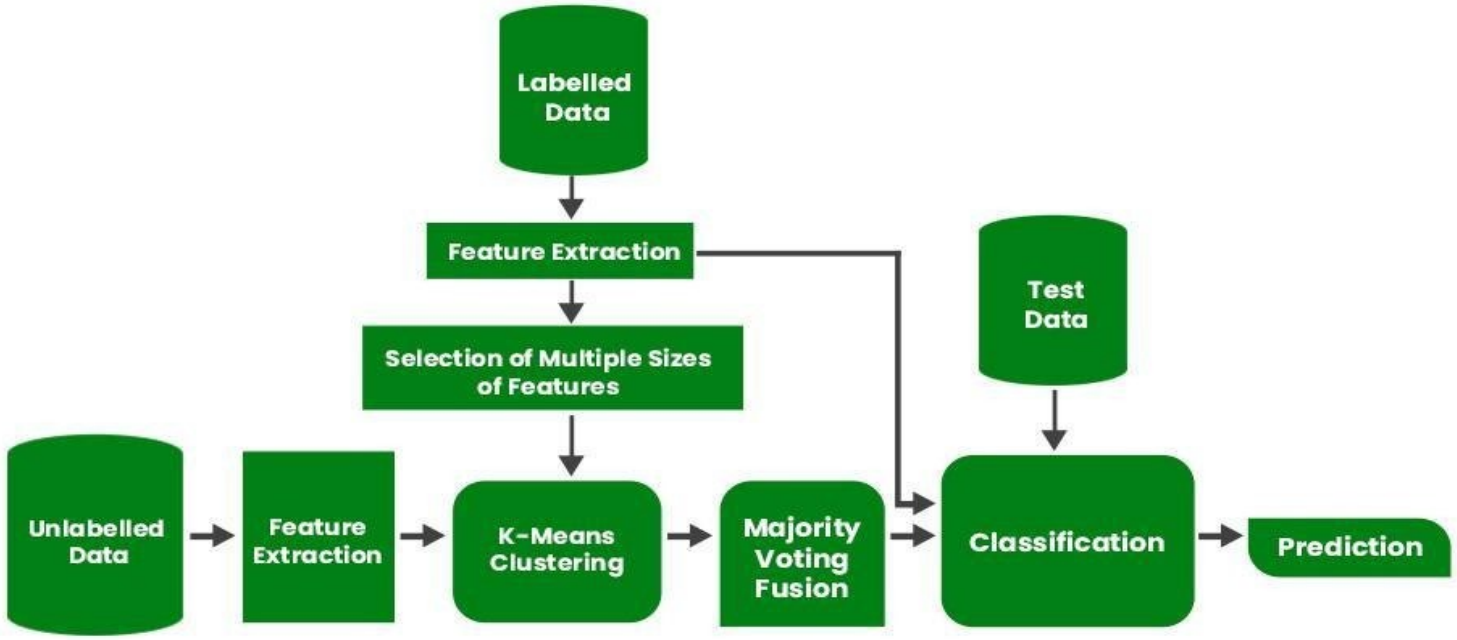


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Semi-Supervised Learning

What is Semi-Supervised Learning?

Semi-supervised learning is a type of machine_learning that falls in between supervised and unsupervised learning. It is a method that uses a small amount of labeled data and a large amount of unlabeled data to train a model.



Text classification:
Image classification:
Anomaly detection:



To overcome the drawbacks of supervised learning and unsupervised learning algorithms, the concept of Semi-supervised learning is introduced.

- The main aim of semi-supervised learning is to effectively use all the available data, rather than only labelled data like in supervised learning.
- Initially, similar data is clustered along with an unsupervised learning algorithm, and further, it helps to label the unlabeled data into labelled data.
- It is because labelled data is a comparatively more expensive acquisition than unlabeled data.
- We can imagine these algorithms with an example.
- Supervised learning is where a student is under the supervision of an instructor at home and college.
- Further, if that student is self - analyzing the same concept without any help from the instructor, it comes under unsupervised learning. Under semi-supervised learning, the student must revise himself after analyzing the same concept under the guidance of an instructor at college.

Advantages and disadvantages of Semi-supervised Learning

Advantages:

- It is simple and easy to understand the algorithm.
- It is highly efficient.
- It is used to solve drawbacks of Supervised and Unsupervised Learning algorithms.

Disadvantages:

- Iterations results may not be stable.
- We cannot apply these algorithms to network-level data.
- Accuracy is low.



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Reinforcement Learning

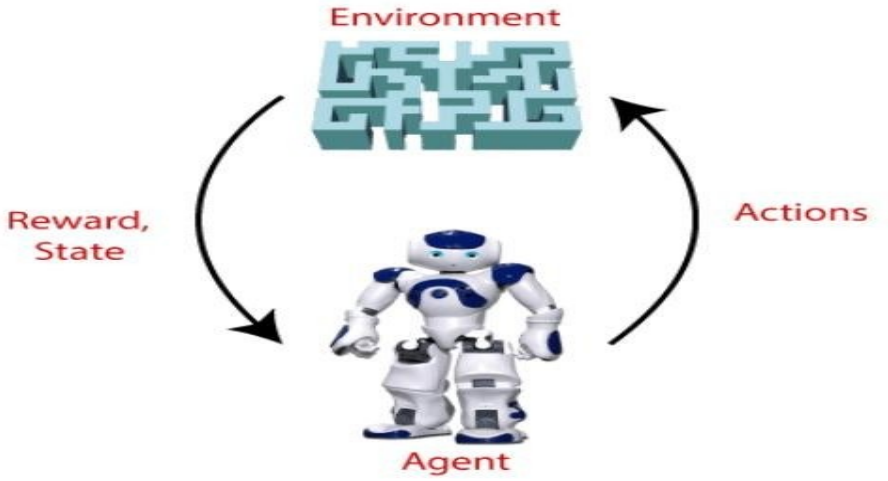
What is Reinforcement Learning?

Reinforcement Learning is a feedback-based Machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions.

For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.

"Reinforcement learning is a type of machine learning method where an intelligent agent (computer program) interacts with the environment and learns to act within that."

The agent continues doing these three things (**take action, change state/remain in the same state, and get feedback**), and by doing these actions, he learns and explores the environment.



Reinforcement Learning

Terms used in Reinforcement Learning

- **Agent():** An entity that can perceive/explore the environment and act upon it.
- **Environment():** A situation in which an agent is present or surrounded by. In RL, we assume the stochastic environment, which means it is random in nature.
- **Action():** Actions are the moves taken by an agent within the environment.
- **State():** State is a situation returned by the environment after each action taken by the agent.
- **Reward():** A feedback returned to the agent from the environment to evaluate the action of the agent.
- **Policy():** Policy is a strategy applied by the agent for the next action based on the current state.
- **Value():** It is expected long-term retuned with the discount factor and opposite to the short-term reward.
- **Q-value():** It is mostly similar to the value, but it takes one additional parameter as a current action (a).

Categories of Reinforcement Learning

- Reinforcement learning is categorized mainly into two types of methods/algorithms:

- **Positive Reinforcement Learning:**

Positive reinforcement learning specifies increasing the tendency that the required behavior would occur again by adding something. It enhances the strength of the behavior of the agent and positively impacts it.

- **Negative Reinforcement Learning:**

Negative reinforcement learning works exactly opposite to the positive RL. It increases the tendency that the specific behavior would occur again by avoiding the negative condition.



Advantages and Disadvantages of Reinforcement Learning

Advantages

•It helps in solving complex real-world problems which are difficult to be solved by general techniques.
•The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.
•Helps in achieving long term results.

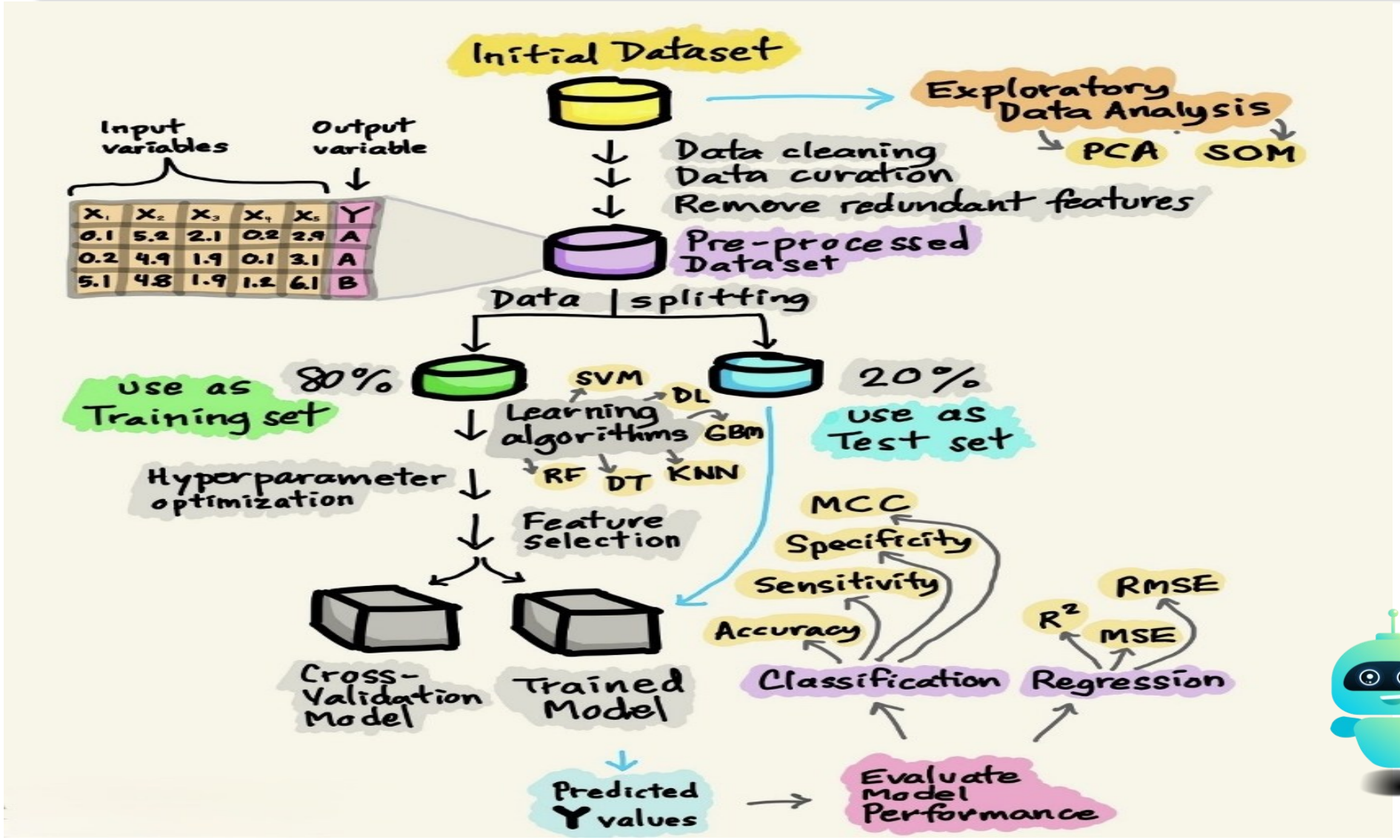
Disadvantage

•RL algorithms are not preferred for simple problems.
•RL algorithms require huge data and computations.
•Too much reinforcement learning can lead to an overload of states which can weaken the results.



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Building Machine Learning Model

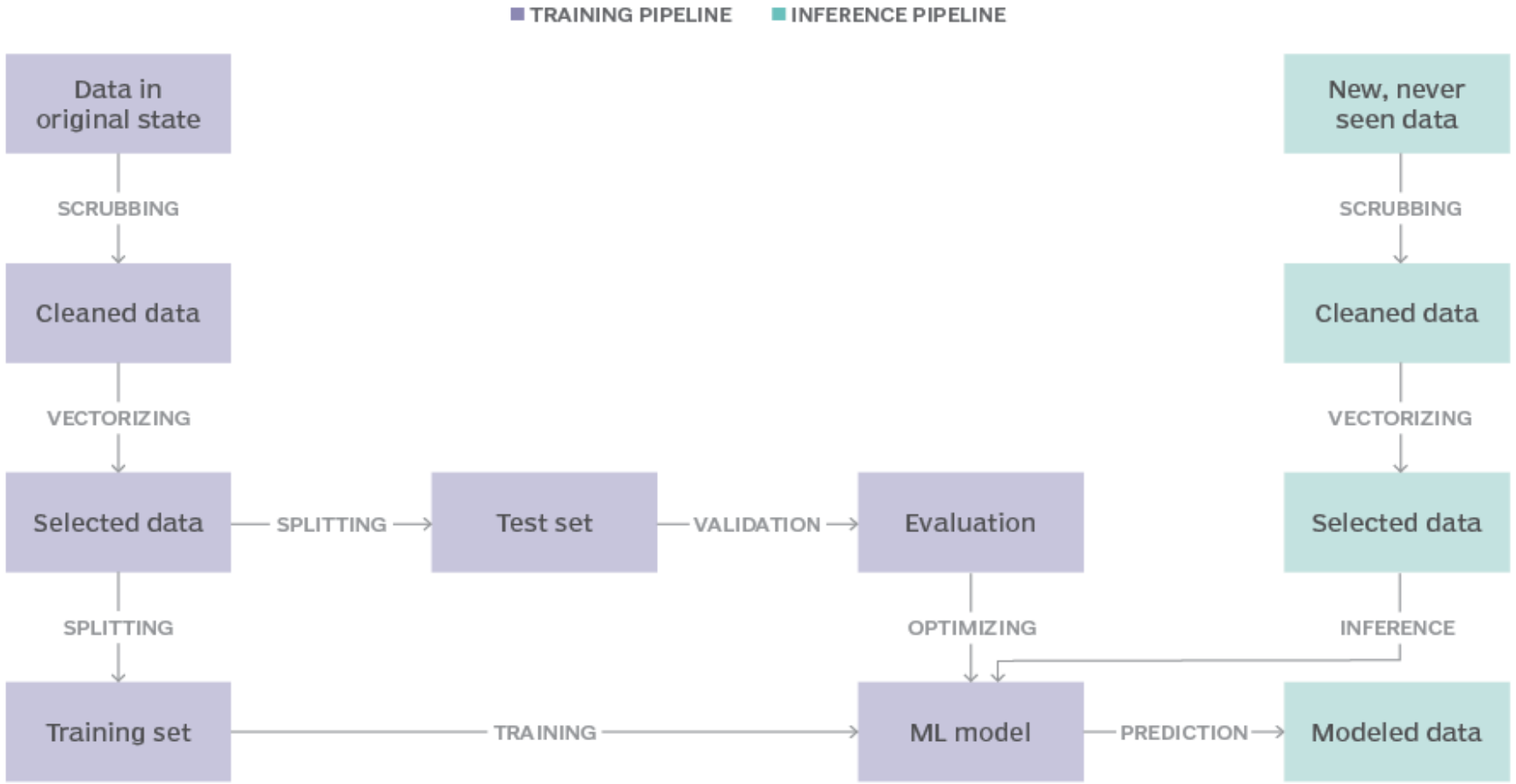


X_1	X_2	X_3	X_4	X_5	Y
0.1	5.2	2.1	0.2	2.9	A
0.2	4.9	1.9	0.1	3.1	A
5.1	4.8	1.9	1.2	6.1	B

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Data Pipeline for Machine Learning

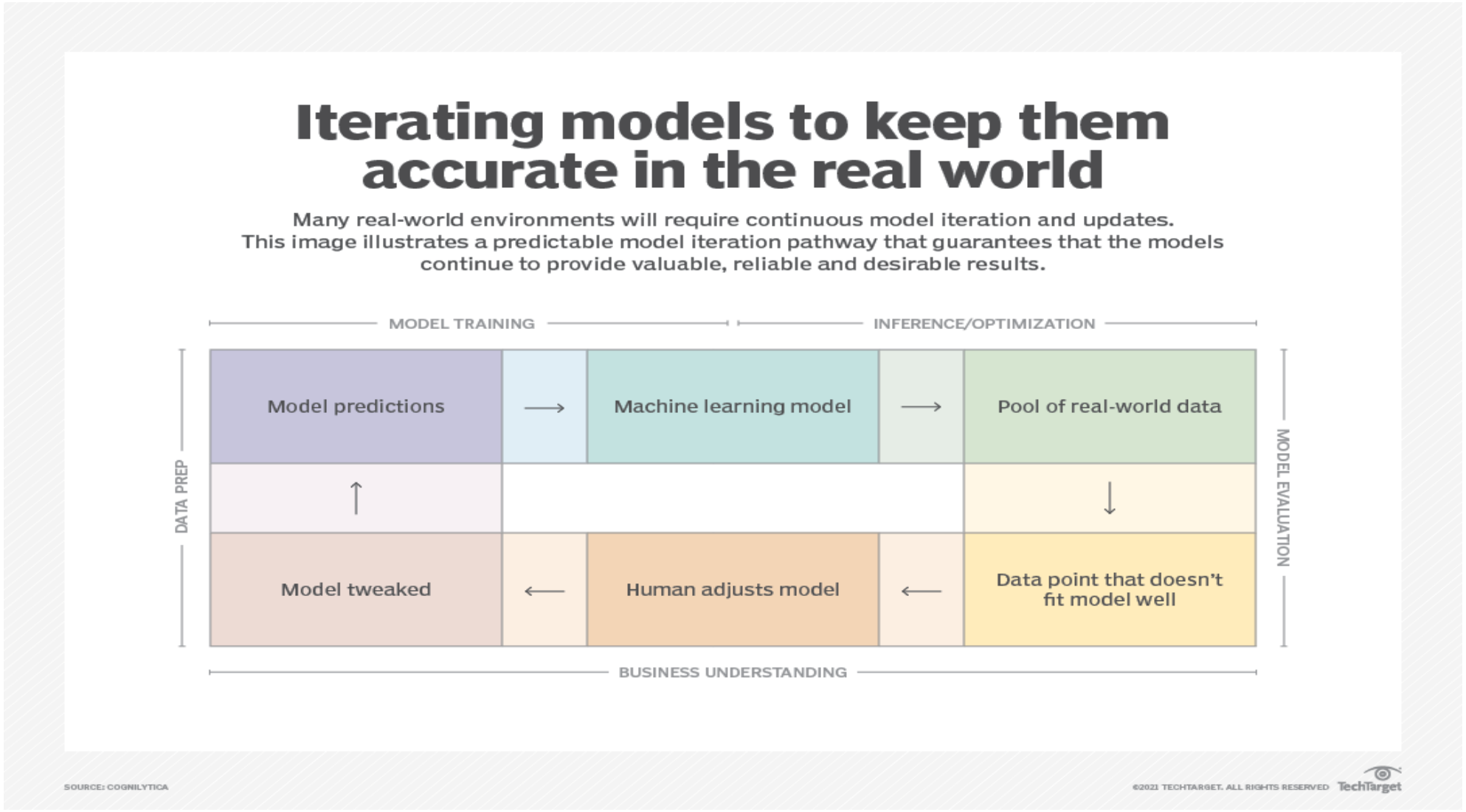
Training pipelines and inference pipelines are both needed in order to continually train machine learning models.





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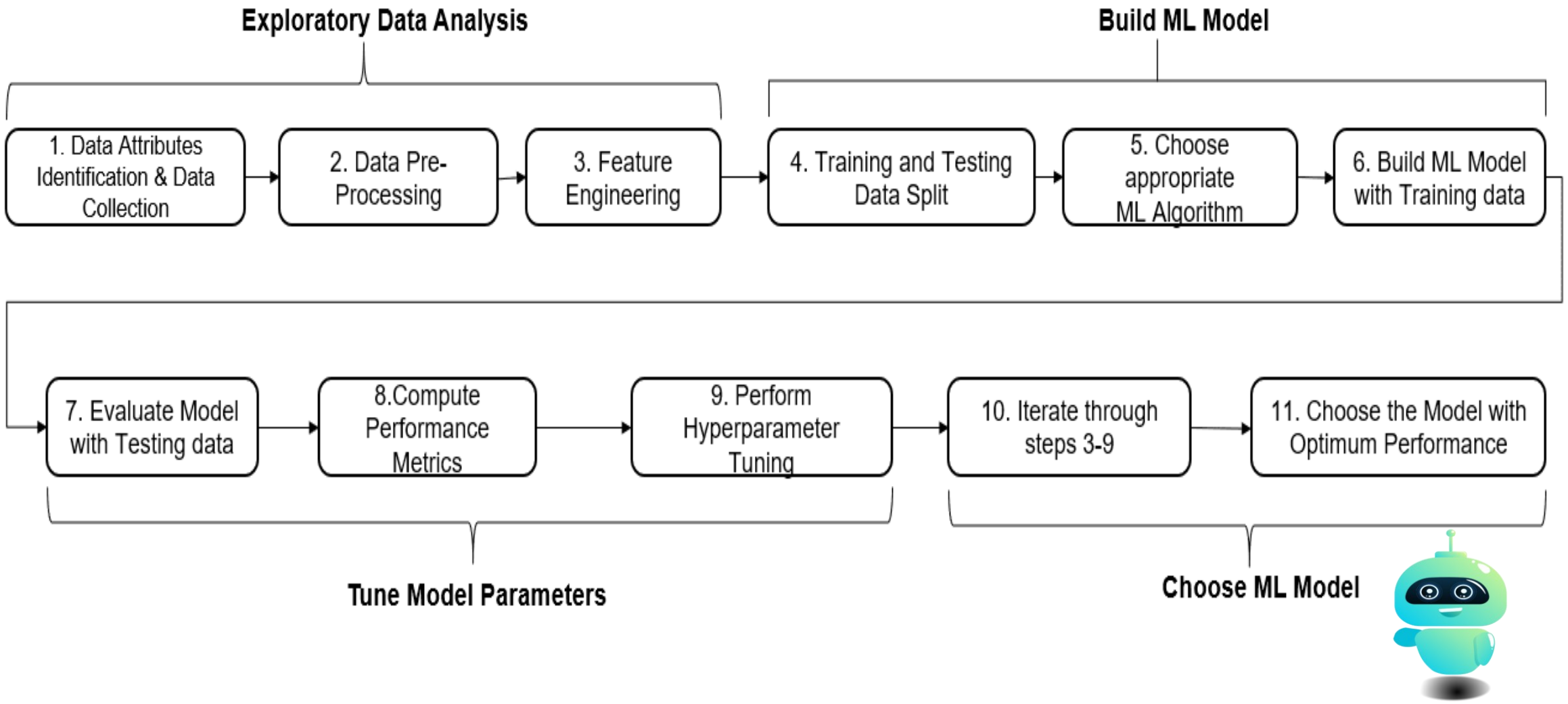
Data Pipeline for Machine Learning





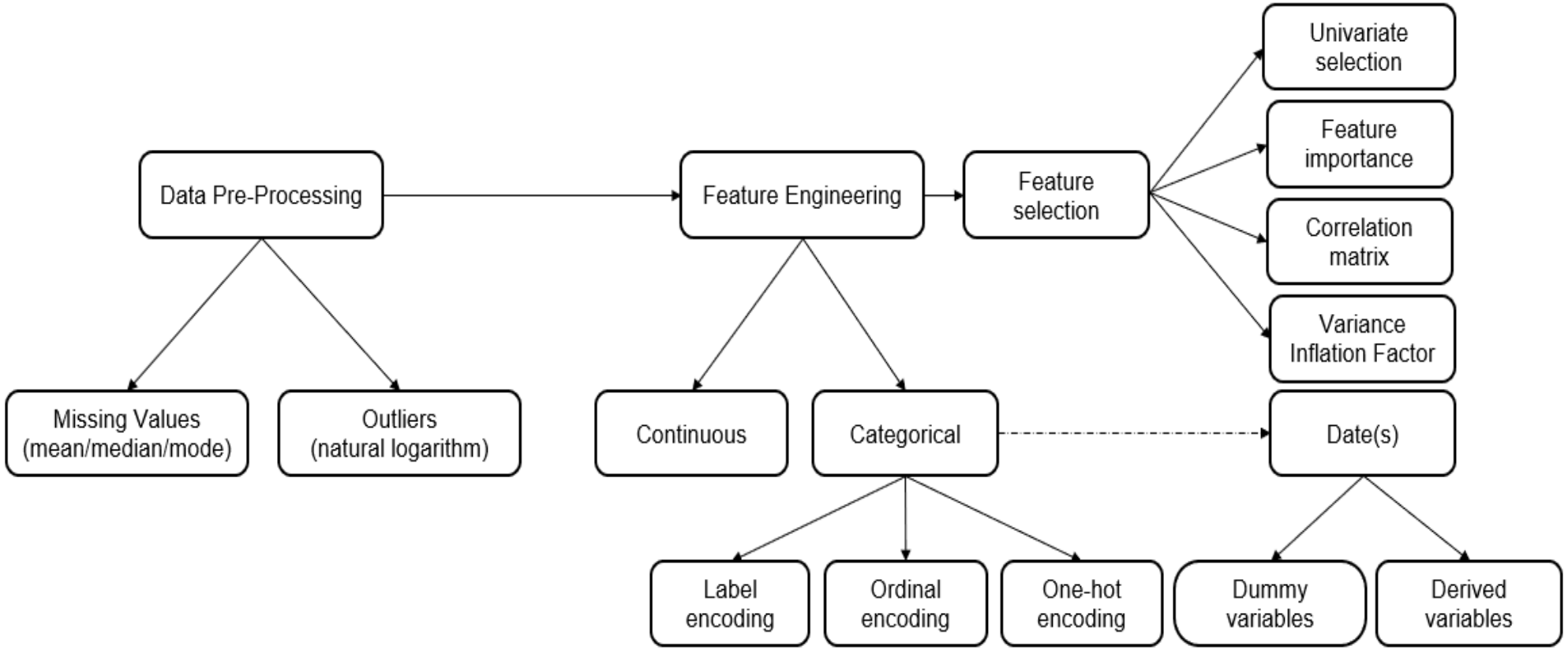
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Data Pipeline for Machine Learning



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Data Pipeline for Machine Learning





Well Defined Learning Problem

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance in tasks T, as measured by P, improves with experience E.

Features in a Learning Problem

- The class of tasks (T)
- The measure of performance to be improved (P)
- The source of experience (E)



Well Defined Learning Problem

Checkers Learning Problem

Task (T)	:	Playing Checkers
Performance Measure (P)	:	Percent of games won against opponents.
Training Experience (E)	:	Playing practice games against itself.

Robot Driving License Problem

Task (T)	:	Driving on public four-lane highways using vision cameras.
Performance Measure (P)	:	Average distance travelled before an error (as judged by a human observer).
Training Experience (E)	:	A sequence of images and steering commands recorded while observing a human driver.

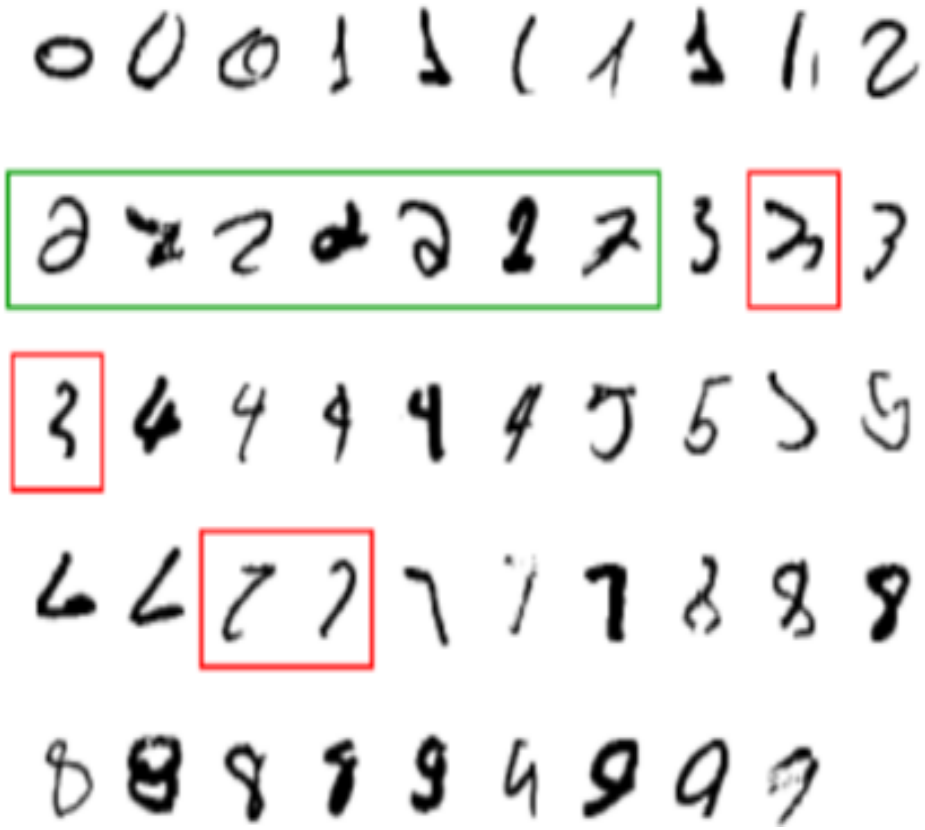


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Well Defined Learning Problem

A classic example of a task that requires machine learning:

It is very hard to say what makes a 2



Handwriting Recognition Learning Problem

Task (T):Recognizing and classifying handwritten words within images.

Performance Measure (P):Percent of words correctly classified.

Training Experience (E): A dataset of handwritten words with given classifications.

https://www.geeksforgeeks.org/well-posed-learning-problems/

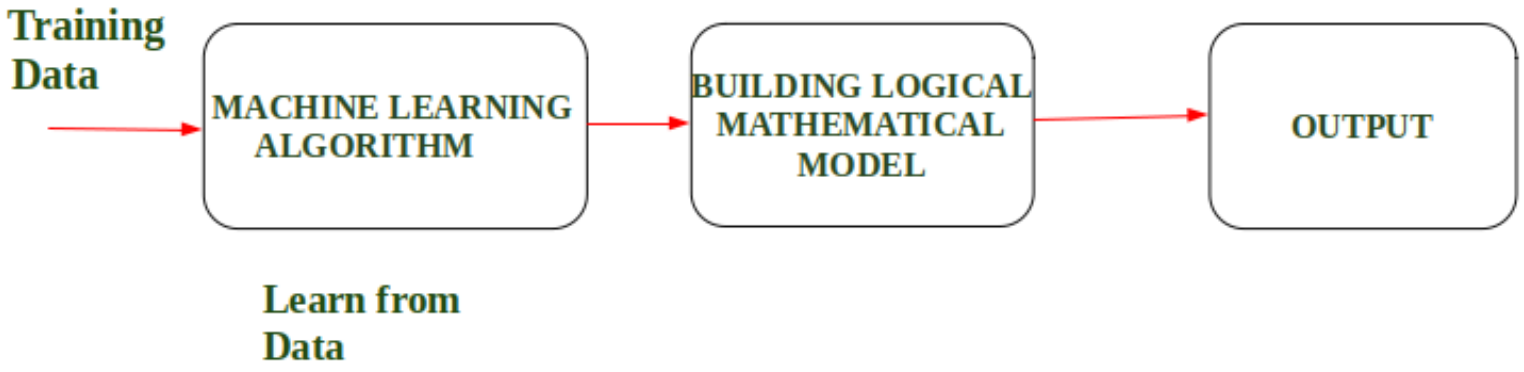


Designing a Learning System

"Machine Learning enables a Machine to Automatically learn from Data, Improve performance from an Experience and predict things without explicitly programmed."

When we fed the Training Data to Machine Learning Algorithm, this algorithm will produce a mathematical model and with the help of the mathematical model, the machine will make a prediction and take a decision without being explicitly programmed.

Also, during training data, the more machine will work with it the more it will get experience and the more efficient result is produced.





Designing a Learning System

Example :

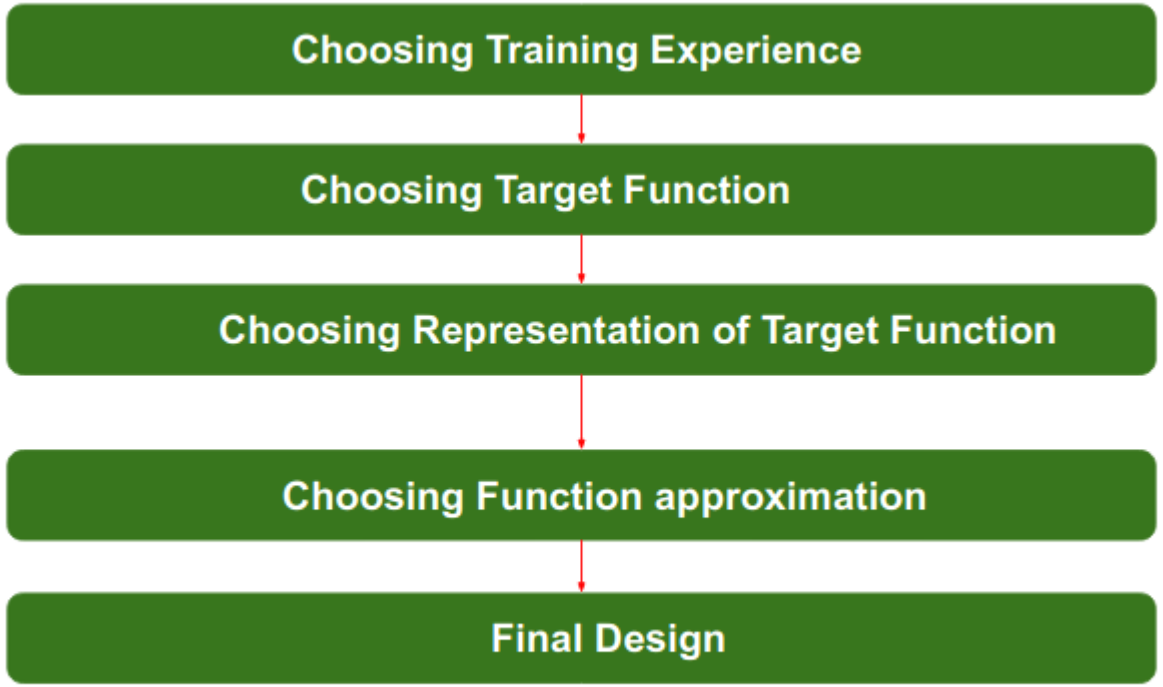
Driverless Car : the training data is fed to Algorithm like how to Drive Car in Highway, Busy and Narrow Street with factors like speed limit, parking, stop at signal etc. After that, a Logical and Mathematical model is created on the basis of that and after that, the car will work according to the logical model. Also, the more data the data is fed the more efficient output is produced.

Example: In Spam E-Mail detection,

- **Task, T:** To classify mails into Spam or Not Spam.
- **Performance measure, P:** Total percent of mails being correctly classified as being "Spam" or "Not Spam".
- **Experience, E:** Set of Mails with label "Spam"



Designing a Learning System

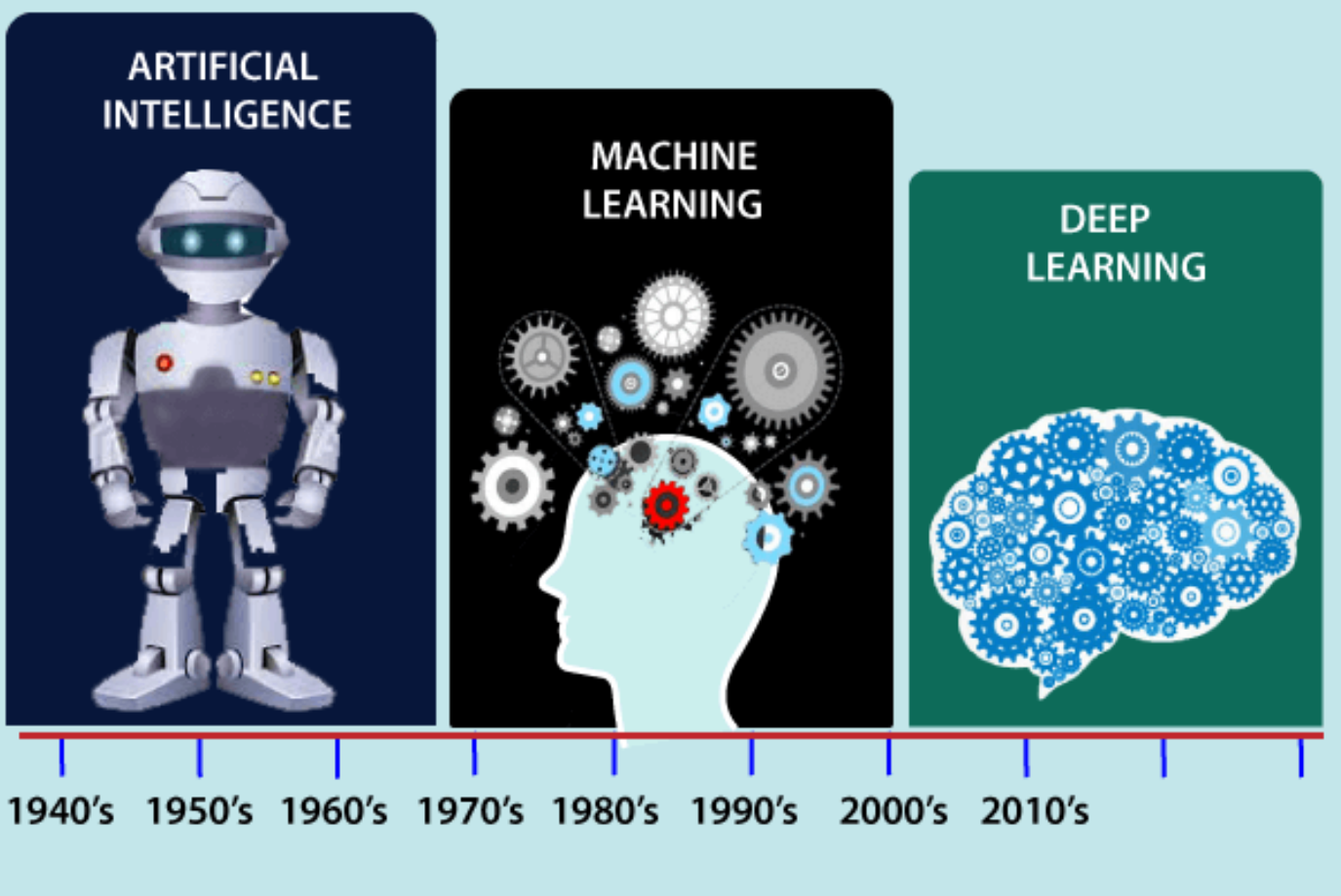


Steps for Designing Learning System are:
https://www.geeksforgeeks.org/design-a-learning-system-in-machine-learning/



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History of Machine Learning



The early history of Machine Learning (Pre-1940):

1834: In 1834, Charles Babbage, the father of the computer, conceived a device that could be programmed with punch cards. However, the machine was never built, but all modern computers rely on its logical structure.

1936: In 1936, Alan Turing gave a theory that how a machine can determine and execute a set of instructions.



History of Machine Learning

The era of stored program computers:

1940: In 1940, the first manually operated computer, "ENIAC" was invented, which was the first electronic general-purpose computer. After that stored program computer such as EDSAC in 1949 and EDVAC in 1951 were invented.

1943: In 1943, a human neural network was modeled with an electrical circuit. In 1950, the scientists started applying their idea to work and analyzed how human neurons might work.

Computer machinery and intelligence:
1950: In 1950, Alan Turing published a seminal paper, "**Computer Machinery and Intelligence**," on the topic of artificial intelligence. **In his paper, he asked, "Can machines think?"**

Machine intelligence in Games:
1952: Arthur Samuel, who was the pioneer of machine learning, created a program that helped an IBM computer to play a checkers game. It performed better more it played.
1959: In 1959, the term "Machine Learning" was first coined by **Arthur Samuel**.

History of Machine Learning

The first "AI" winter:

The duration of 1974 to 1980 was the tough time for AI and ML researchers, and this duration was called as **AI winter**.

In this duration, failure of machine translation occurred, and people had reduced their interest from AI, which led to reduced funding by the government to the researches.

Machine Learning from theory to reality

1959: In 1959, the first neural network was applied to a real-world problem to remove echoes over phone lines using an adaptive filter.

1985: In 1985, Terry Sejnowski and Charles Rosenberg invented a neural network **NETtalk**, which was able to teach itself how to correctly pronounce 20,000 words in one week.

1997: The IBM's **Deep blue** intelligent computer won the chess game against the chess expert Garry Kasparov, and it became the first computer which had beaten a human chess expert.

History of Machine Learning

Machine Learning at 21st century

2006: In the year 2006, computer scientist Geoffrey Hinton has given a new name to neural net research as "**deep learning**," and nowadays, it has become one of the most trending technologies.

2012: In 2012, Google created a deep neural network which learned to recognize the image of humans and cats in YouTube videos.

2014: In 2014, the Chabot "**Eugen Goostman**" cleared the Turing Test. It was the first Chabot who convinced the 33% of human judges that it was not a machine.

2014: DeepFace was a deep neural network created by Facebook, and they claimed that it could recognize a person with the same precision as a human can do.

2016: AlphaGo beat the world's number second player **Lee sedol** at **Go game**. In 2017 it beat the number one player of this game **Ke Jie**.

2017: In 2017, the Alphabet's Jigsaw team built an intelligent system that was able to learn the **online trolling**. It used to read millions of comments of different websites to learn to stop online trolling.

History of Machine Learning

Machine Learning at present:

Now machine learning has got a great advancement in its research, and it is present everywhere around us, such as **self-driving cars**, **Amazon Alexa**, **CatBoots**, **recommender system**, and many more. It includes **Supervised**, **unsupervised**, and **reinforcement learning with clustering**, **classification**, **decision tree**, **SVM algorithms**, etc.

Modern machine learning models can be used for making various predictions, including **weather prediction**, **disease prediction**, **stock market analysis**, etc.



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Introduction to Machine Learning Approaches

- A machine learning approach refers to the algorithms, techniques, and methods used to create models that solve real-world problems by processing data.
- These approaches can be categorized based on the amount of human supervision, the ability to learn from incremental data samples, and their approach to generalization from data samples.
- Machine learning has numerous algorithms, techniques, and methods that can be utilized to create models to solve real-world problems employing data.

Supervised learning
Unsupervised learning
Semi-supervised learning
Reinforcement Learning
Neural network
Decision tree

Support Vector machines
Naive Bayes
Compute cluster
K-nearest neighbors
Anomaly detection
Logistic regression

Random Forest
Natural language processing
Classification
Computer vision
Linear models
Regression



Introduction to Machine Learning Approaches

neural network

Neural Networks are computational models that mimic the complex functions of the human brain.

neural networks consist of interconnected nodes or neurons that process and learn from data, enabling tasks such as pattern recognition and decision making in machine learning.

1940s-1950s: Early Concepts

1960s-1970s: Perceptron

1980s: Backpropagation and Connectionism

1990s: Boom and Winter

2000s: Resurgence and Deep Learning

2010s-Present: Deep Learning Dominance



Introduction to Machine Learning Approaches

neural network

A neural network is an artificial system made of interconnected nodes (neurons) that process information, modeled after the structure of the human brain. It is employed in machine learning jobs where patterns are extracted from data.

Layers of connected neurons process data in neural networks. The network processes input data, modifies weights during training, and produces an output depending on patterns that it has discovered.

Feedforward neural networks
recurrent neural networks (RNNs)
convolutional neural networks (CNNs) and
long short-term memory networks (LSTMs) are examples of common architectures that are each designed for a certain task.

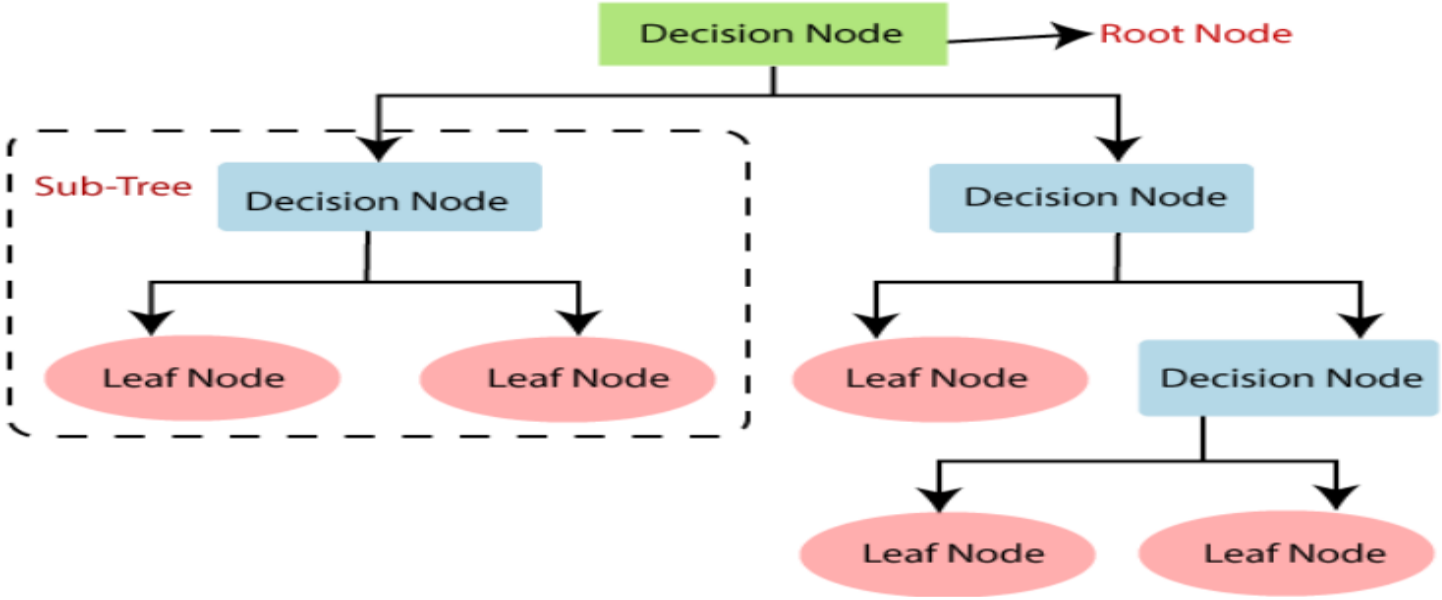
Introduction to Machine Learning Approaches

Decision tree

Decision Tree is a **Supervised learning technique** that can be used for both classification and Regression problems

Decision Node and **Leaf Node.** Decision nodes are used to make any decision and have multiple branches

It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.





Introduction to Machine Learning Approaches

Support Vector machines

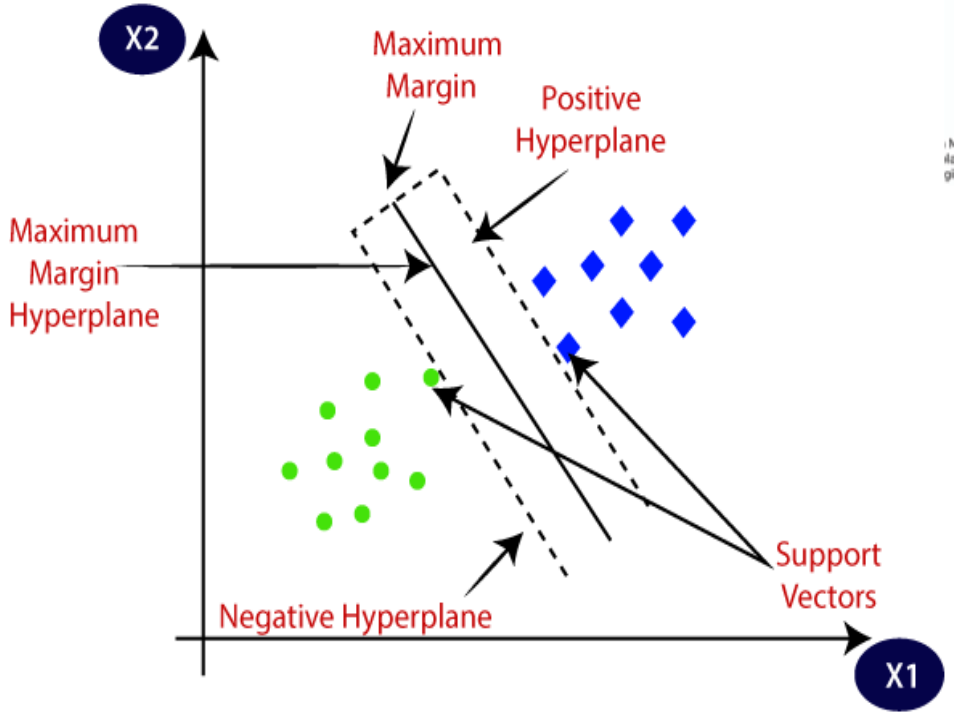
It is the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes

Then we can easily put the new data point in the correct category in the future.

This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane.

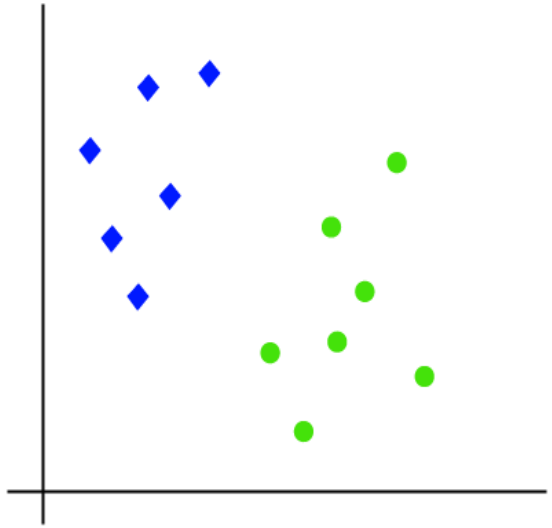


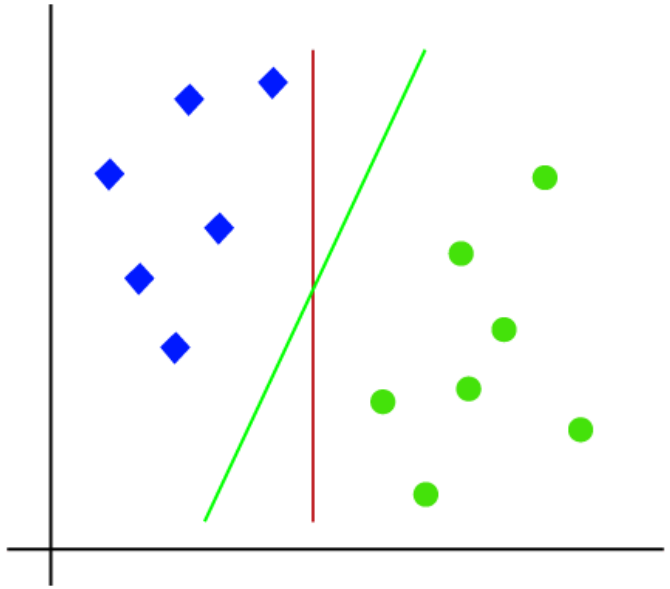
Introduction to Machine Learning Approaches

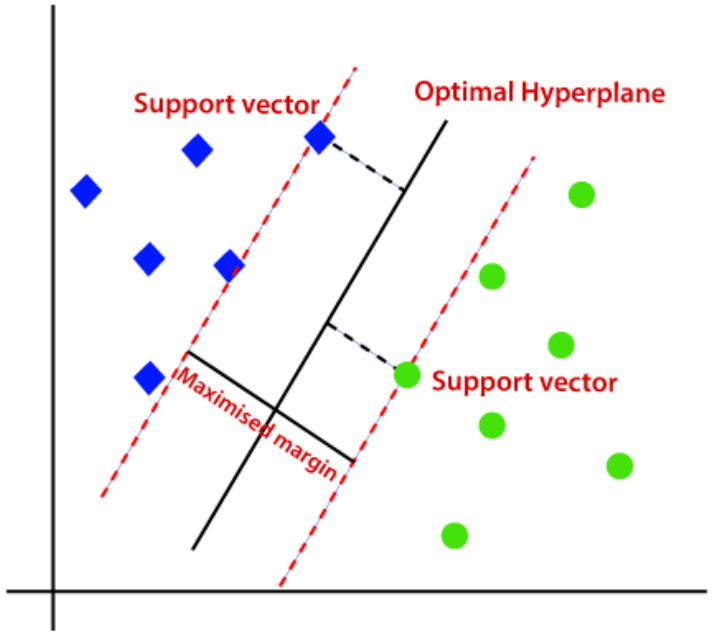
Support Vector machines

A **hyperplane** is a decision boundary that divides the input space into two or more regions, each corresponding to a different class or output label.

Linear







Introduction to Machine Learning Approaches

Naïve Bayes theorem

Naïve Bayes algorithm is a supervised learning algorithm, which is based on **Bayes theorem** and used for solving classification problems.

It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.

Some popular examples of Naïve Bayes Algorithm are **spam filtration, Sentimental analysis, and classifying articles**.

Bayes' Theorem:

•Bayes' theorem is also known as **Bayes' Rule** or **Bayes' law**, which is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.

•The formula for Bayes' theorem is given as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



https://www.javatpoint.com/machine-learning-naive-bayes-classifier

Introduction to Machine Learning Approaches

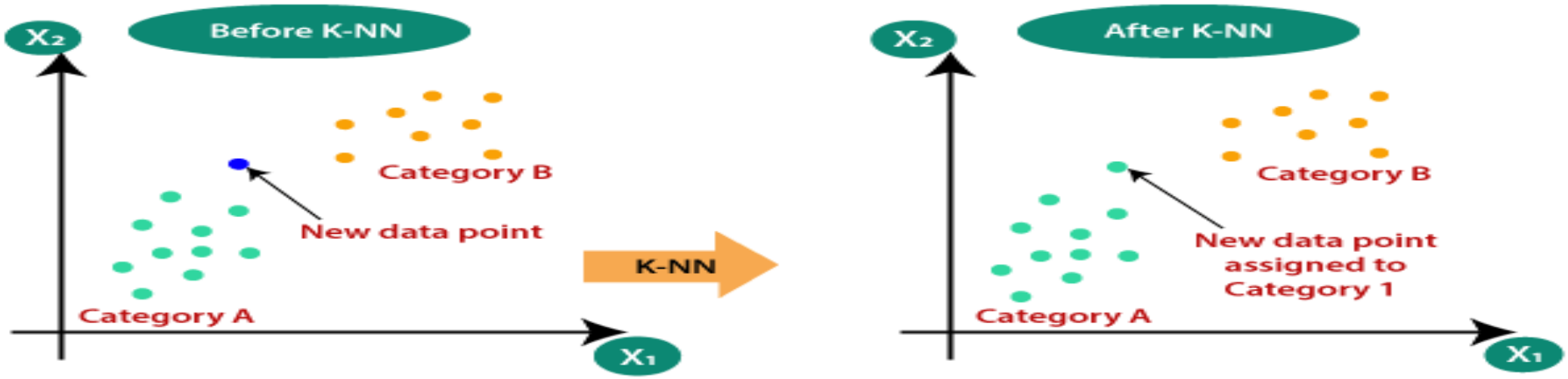
The **K-Nearest Neighbors (KNN) algorithm** is a supervised machine learning method employed to tackle classification and regression problems.

K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.

K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.

K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

Introduction to Machine Learning Approaches



How does K-NN work?

The K-NN working can be explained on the basis of the below algorithm:

- **Step-1:** Select the number K of the neighbors
- **Step-2:** Calculate the Euclidean distance of **K number of neighbors**
- **Step-3:** Take the K nearest neighbors as per the calculated Euclidean distance.
- **Step-4:** Among these k neighbors, count the number of the data points in each category.
- **Step-5:** Assign the new data points to that category for which the number of the neighbor is maximum.
- **Step-6:** Our model is ready.



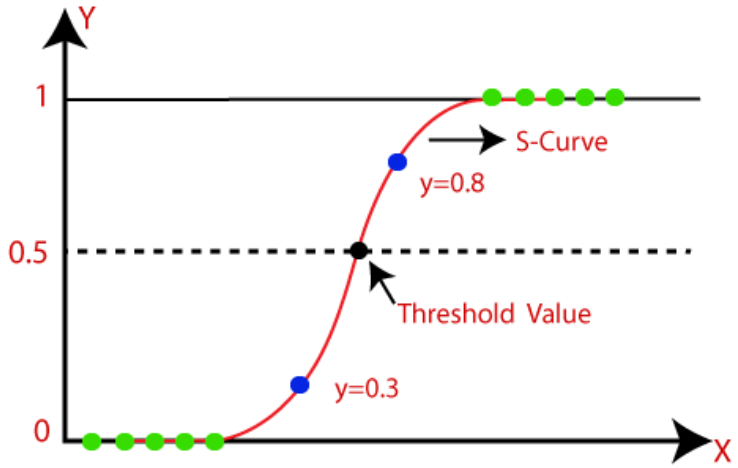
Introduction to Machine Learning Approaches

Logistic Regression

Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value.

It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1**.

Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas **Logistic regression is used for solving the classification problems**.



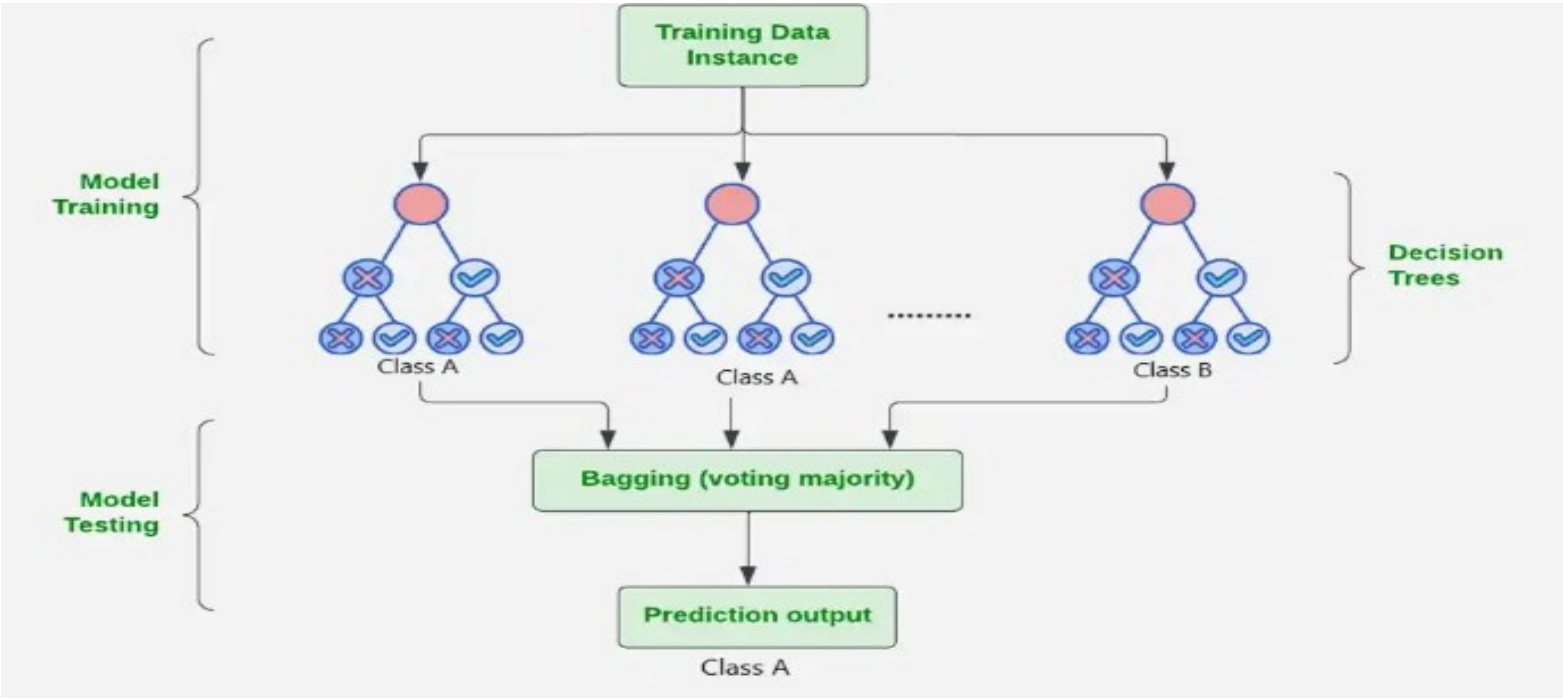


Introduction to Machine Learning Approaches

Random Forest Algorithm

It works by creating a number of Decision Trees during the training phase.

Each tree is constructed using a random subset of the data set to measure a random subset of features in each partition.

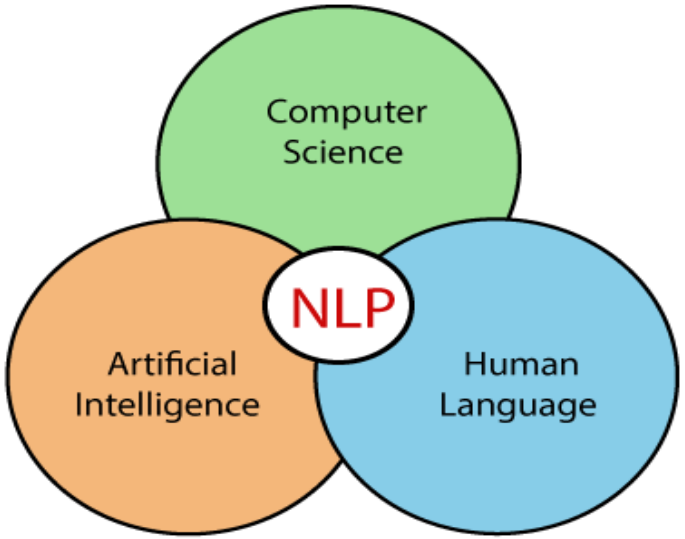




Introduction to Machine Learning Approaches

What is NLP?

NLP stands for **Natural Language Processing**, which is a part of **Computer Science, Human language,** and **Artificial Intelligence**. It is the technology that is used by machines to understand, analyse, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as **translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction,** and **topic segmentation**.

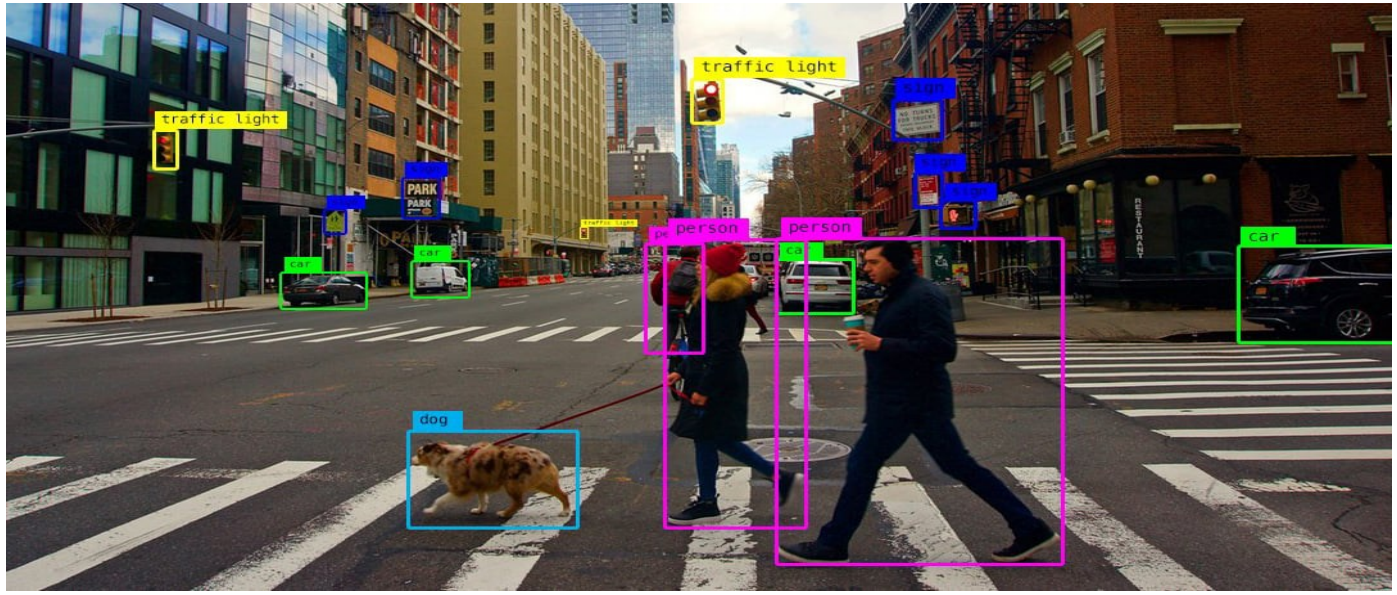




Introduction to Machine Learning Approaches

What is Computer Vision?

Computer vision is a field of study within artificial intelligence (AI) that focuses on enabling computers to Intercept and extract information from images and videos, in a manner similar to human vision. It involves developing algorithms and techniques to extract meaningful information from visual inputs and make sense of the visual world.





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Features and Their Types

What is a Feature?

A feature (also called an attribute, predictor, or independent variable) is an individual measurable property or characteristic of a data point. Features are the INPUTS to a ML model. Choosing and engineering good features is often more important than the choice of algorithm.

Example Dataset — House Price Prediction

House ID	Area (sqft)	Bedrooms	Location	Age (yrs)	Price (₹L)
1	1200	2	Noida	5	75
2	1800	3	Gurgaon	2	150
3	950	1	Delhi	12	55

Features (X): `Area, Bedrooms, Location, Age` → **Target (y):** `Price`

Feature Type Categories

- Numerical
- Categorical
- Text
- Date / Time

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Numerical Features

Definition

Numerical (quantitative) features represent measurable quantities with a meaningful magnitude. They support arithmetic operations (addition, subtraction, ratios) and are most commonly used as direct inputs to ML models.

Continuous Numerical

Can take any real value within a range.

Examples:

- Temperature (23.5 °C)
- Height (174.6 cm)
- Salary (₹65,234.50)
- Distance, weight, time

Distribution: usually plotted as a histogram or KDE; often normally distributed after transformation.

Models that use directly:

Linear / Logistic Regression, Neural Networks, SVMs.

Discrete Numerical

Integer values only — countable.

Examples:

- Number of children (0, 1, 2, ...)
- Bedrooms in a house
- Customer purchase count
- Page views per session

Distribution: often modelled with Poisson, Binomial, or Negative Binomial distributions.

Caveat:

May need treatment as categorical if cardinality is low (e.g. ratings 1-5).

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Categorical Features

Definition

Categorical (qualitative) features take values from a limited set of discrete CATEGORIES. They describe labels, groups, or types — not magnitudes. Most ML algorithms cannot use them directly; they need encoding to numbers first.

Nominal (Unordered)

Categories have NO natural order.

Examples:

- Gender: {Male, Female, Other}
- Country: {India, US, UK, Japan, ...}
- Colour: {Red, Green, Blue}
- Blood type: {A, B, AB, O}

Encoding:

One-Hot Encoding (best default)
Target / Frequency encoding (high cardinality)

Avoid: *Label-encoding nominal features — creates fake ordering.*

Ordinal (Ordered)

Categories HAVE a meaningful order.

Examples:

- Education: {HS < Bachelor < Master < PhD}
- T-shirt size: {S < M < L < XL}
- Rating: {Poor < Avg < Good < Excellent}
- Customer tier: {Bronze < Silver < Gold}

Encoding:

Ordinal encoding (preserves order)

```
e.g. {S, M, L, XL} → {1, 2, 3, 4}
```

Caveat: *Gaps between levels are assumed equal — may not always be true.*

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Other Feature Types & Engineering

Beyond Numerical & Categorical

Text

Free text from reviews, emails, articles.
Must be vectorized (BoW, TF-IDF, embeddings).

Image

Pixel arrays. Used directly by CNNs or via pre-extracted embeddings.

Date / Time

Timestamps. Engineer hour, day-of-week, season, time-since-event.

Derived / Engineered

Combinations of base features: BMI = weight / height², log(price), ratios.

Worked Example — Bank Customer Churn Dataset

Feature	Example Values	Type
credit_score	350, 620, 750, 850	Numerical (Continuous)
age	23, 35, 47, 65	Numerical (Discrete)
geography	France, Spain, Germany	Categorical (Nominal)
card_tier	Silver, Gold, Platinum	Categorical (Ordinal)
account_open_date	2018-04-12	Date / Time
complaint_text	"poor service experience"	Text

Identifying feature types CORRECTLY drives every downstream preprocessing choice.

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Handling Missing Data

Why Missing Data Matters

Real-world datasets often have missing values — sensor failures, survey skips, data-entry errors. Most ML algorithms cannot handle NaN/NULL directly. Ignoring or mishandling missing data introduces bias and degrades model performance significantly.

Three Types of Missingness

MCAR

Missing Completely At Random

Missingness is unrelated to any value, observed or unobserved.

Example:

A few survey forms accidentally torn during transport.

Fix:

Safe to drop or simple-impute.

MAR

Missing At Random

Missingness depends on OBSERVED features but not the missing value itself.

Example:

Younger users skip the income field more often than older users.

Fix:

Use model-based imputation conditioning on observed features.

MNAR

Missing Not At Random

Missingness depends on the missing value itself — most problematic.

Example:

High earners are less likely to report their income.

Fix:

Hardest to handle — needs explicit modelling of the missingness mechanism.

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Imputation Techniques

1. Delete (Listwise / Pairwise)

Drop rows or columns with missing values. Simple but wastes data; safe only if missingness is small (< 5%) and MCAR.

2. Mean / Median / Mode Imputation

Fill with feature's mean (numerical), median (skewed numerical), or mode (categorical). Fast, but reduces variance and ignores correlations.

3. Forward / Backward Fill

Use the previous (or next) value in a time-series. Useful for sensor data with brief gaps.

4. K-Nearest Neighbors Imputation

Find K most similar rows in feature space; fill with their average. Preserves relationships.

5. Model-Based (Regression / Random Forest)

Train a model with non-missing features predicting the missing one. Powerful but adds complexity.

6. MICE (Multiple Imputation by Chained Equations)

Iteratively impute each missing feature using regressions on others; combines multiple imputations for robust uncertainty estimates.

Missing Data — Code Example

Common techniques in pandas + scikit-learn:

```
import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer, KNNImputer
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer

df = pd.read_csv("customers.csv")

# Inspect missingness
print(df.isnull().sum())                  # count per column
print(df.isnull().mean() * 100)           # percentage missing

# 1. Mean imputation for numerical
df["age"] = df["age"].fillna(df["age"].mean())

# 2. Mode imputation for categorical
df["city"] = df["city"].fillna(df["city"].mode()[0])

# 3. KNN imputation (preserves relationships)
imp = KNNImputer(n_neighbors=5)
df_filled = pd.DataFrame(imp.fit_transform(df.select_dtypes(np.number)),
                         columns=df.select_dtypes(np.number).columns)

# 4. MICE – iterative multivariate imputation
imp = IterativeImputer(random_state=42)
df_mice = imp.fit_transform(df.select_dtypes(np.number))
```

Dealing with Categorical Features

Why Encode?

Most ML algorithms work only with numerical inputs. Categorical values like "Red", "Blue", "Green" must be converted to numbers — but HOW we encode them affects what the model learns. The right encoding depends on cardinality, ordinality, and the algorithm in use.

Six Common Encoding Methods

Label Encoding

Assign integer to each category

One-Hot Encoding

Binary column per category

Ordinal Encoding

Integer mapping respecting order

Target Encoding

Replace by mean target value

Binary Encoding

Binary representation of integers

Frequency Encoding

Replace by category frequency

Quick Decision Guide

- **Nominal, low cardinality (≤ 15 categories):** One-Hot Encoding.
- **Ordinal:** Ordinal encoding (preserves order).
- **Nominal, high cardinality (> 50 categories):** Target / Frequency / Binary encoding.
- **Tree-based models (RF, XGBoost):** Often work with Label encoding directly.

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Label Encoding

Definition

Label encoding assigns a unique INTEGER to each category — usually alphabetically or by first appearance. Simple and memory-efficient, but introduces an artificial ordering between categories.

Example

Original	Label Encoded
Red	0
Blue	1
Green	2
Red	0

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df["color_enc"] = le.fit_transform(df["color"])

# OR using pandas:
df["color_enc"] =
df["color"].astype("category").cat.codes
```

Use when

- Categories have a natural ORDER.
- Target variable is being encoded (sklearn convention).
- Working with tree-based models that split on any threshold.

Avoid when

- Categories are NOMINAL (unordered).
- Using linear or distance-based models — they'll treat 0, 1, 2 as ordered numerical.
- Risk: model may infer that Green > Blue > Red — a meaningless ordering.

One-Hot Encoding

Definition

One-Hot encoding creates a separate BINARY column for EACH category. The column for that category is 1, all others are 0. No artificial ordering is introduced — the gold standard for nominal features with low cardinality.

Example

color	color_Red	color_Blue	color_Green
Red	**1**	0	0
Blue	0	**1**	0
Green	0	0	**1**

Python Code

```
# Option 1 – pandas (simple)
df = pd.get_dummies(df, columns=["color"], drop_first=True)
#   drop_first=True avoids the "dummy variable trap" (multicollinearity)

# Option 2 – sklearn (production pipelines)
from sklearn.preprocessing import OneHotEncoder
enc = OneHotEncoder(sparse_output=False, drop="first", handle_unknown="ignore")
X_enc = enc.fit_transform(df[["color"]])
# fit on train, .transform on test → consistent columns across splits.
```

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Target, Ordinal & Other Encodings

Ordinal Encoding

Manually map categories to integers RESPECTING their natural order.

```
size_map = {"Small": 1, "Medium": 2, "Large": 3, "XL": 4}
df["size_enc"] = df["size"].map(size_map)
```

Best for: education level, T-shirt size, customer tier, rating scales.

Target / Mean Encoding

Replace each category with the MEAN of the target variable for that category. Powerful for high-cardinality features.

```
city_avg_price = df.groupby("city")["price"].mean()
df["city_enc"] = df["city"].map(city_avg_price)
```

Caveat: *leakage risk! Compute on training set only, apply to test. Use cross-fold within training to avoid overfitting.*

Binary & Frequency Encoding (for high-cardinality)

Binary Encoding: Label-encode, then convert to binary representation — only $[\log_2 N]$ columns instead of N (one-hot). Saves memory when N is large.
Frequency Encoding: Replace each category with its frequency (count) in training data — captures category popularity.

```
df["city_freq"] = df["city"].map(df["city"].value_counts())
```

Best for: zip codes, product IDs, user IDs — anything with hundreds of unique values.

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Feature Scaling

Why Scale Features?

In real datasets, features have wildly different RANGES. Age might be 0-100; salary might be 10,000-10,000,000. Algorithms that compute distances or use gradients treat the larger-scale feature as more important — even when it isn't. Scaling brings features to comparable ranges.

Algorithms NEEDING Scaling

Distance-based:

- k-Nearest Neighbors
- K-Means clustering
- Support Vector Machines (RBF)

Gradient-based:

- Linear / Logistic Regression
- Neural Networks

Component analysis:

- PCA (driven by variance)
- LDA

Regularization:

- Ridge / LASSO Regression

Algorithms NOT Needing Scaling

Tree-based models split on threshold:

- Decision Trees
- Random Forest
- Gradient Boosting (XGBoost, LightGBM)

Probabilistic:

- Naive Bayes (operates on probabilities)

Why?

These models look at FEATURE RANKINGS or thresholds, not magnitudes. Scaling doesn't affect splits.

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Min-Max Normalization vs Standardization

Min-Max Normalization

$$x' = (x - x_min) / (x_max - x_min)$$

Output range: `[0, 1]`
Use when:

- Bounded data is needed (e.g. image pixels)
- You know the min and max
- Data is NOT normally distributed

Standardization (Z-score)

$$x' = (x - \mu) / \sigma$$

Output: `mean=0, std=1 (no fixed bound)`
Use when:

- Data is normally distributed
- Outliers are present (more robust)
- PCA, SVMs, gradient-based models

Worked Example

Original ages: [25, 30, 35, 40, 45] → min=25, max=45, μ=35, σ≈7.07

Min-Max scaled: [0.00, 0.25, 0.50, 0.75, 1.00]
Standardized: [-1.41, -0.71, 0.00, 0.71, 1.41]
Both transformations preserve **RELATIVE** *differences — just rescale the values.*

Feature Scaling — Code Example

Apply scaling using scikit-learn — ALWAYS fit on training data, then transform both train and test:

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler, RobustScaler
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Min-Max scaling — output in [0, 1]
mms = MinMaxScaler()
X_train_mm = mms.fit_transform(X_train)      # fit + transform on TRAIN
X_test_mm  = mms.transform(X_test)            # only transform on TEST

# Standardization — mean=0, std=1
ss = StandardScaler()
X_train_ss = ss.fit_transform(X_train)
X_test_ss  = ss.transform(X_test)

# RobustScaler — uses median and IQR; better when outliers exist
rs = RobustScaler()
X_train_rs = rs.fit_transform(X_train)
```

Cardinal Rule

Never call fit() on the test set — that causes DATA LEAKAGE. Always fit the scaler on training data ONLY, then apply (transform) the same scaler to test/validation/production data. This ensures consistent and unbiased evaluation.

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Feature Selection

What is Feature Selection?

Feature selection is the process of choosing the most RELEVANT subset of features for model training — and discarding noisy, redundant, or irrelevant ones. Different from feature ENGINEERING (creating new features). Different from dimensionality reduction (PCA — creates linear combinations).

Why Select Features?

Avoid Curse of Dimensionality

As features grow, data becomes sparse. Models need exponentially more data to generalize.

Reduce Overfitting

Fewer features = simpler model = less variance. Removes noise that mimics signal.

Faster Training

Less data to process, less memory used. Critical for production systems.

Improve Interpretability

Easier to explain a model using 5 features than 500.

The Curse of Dimensionality (Bellman, 1957)

As feature count grows, the volume of feature space grows EXPONENTIALLY. Data points become equidistant from each other — distance-based methods (k-NN, k-means) lose meaning. To maintain density, you need 10^d training points for d features. Fix: select features carefully or use dimensionality reduction.

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Feature Selection Approaches

Filter Methods

Use statistical tests INDEPENDENT of any ML model. Rank each feature by its relationship with the target.

Examples:

- *Chi-Square (categorical → categorical)*
- *Pearson / Spearman correlation*
- *ANOVA F-test*
- *Mutual Information*

Pros:

Fast, model-agnostic, no overfitting risk.

Cons:

Ignores feature interactions.

Wrapper Methods

Treat feature selection as a search problem. Train MANY models with different feature subsets and pick the best.

Examples:

- *Forward Selection (add one feature at a time)*
- *Backward Elimination (drop one at a time)*
- *Recursive Feature Elimination (RFE)*

Pros:

Captures feature interactions; best accuracy.

Cons:

Computationally expensive; risk of overfitting.

Embedded Methods

Selection happens DURING model training. The algorithm naturally identifies important features.

Examples:

- *LASSO regression (L1 penalty drives coefficients to 0)*
- *Tree-based feature importance (RF, XGBoost)*
- *Elastic Net*

Pros:

Good balance of speed and accuracy.

Cons:

Tied to specific model family.

Feature Selection — Code Example

Three practical techniques side-by-side:

```
from sklearn.feature_selection import SelectKBest, chi2, RFE
from sklearn.linear_model import LogisticRegression, Lasso
from sklearn.ensemble import RandomForestClassifier

# Filter – Chi-square on categorical features
selector = SelectKBest(score_func=chi2, k=10)
X_new = selector.fit_transform(X, y)
print("Selected features:", X.columns[selector.get_support()])

# Wrapper – Recursive Feature Elimination
model = LogisticRegression(max_iter=1000)
rfe = RFE(estimator=model, n_features_to_select=10)
X_rfe = rfe.fit_transform(X, y)
print("Ranking:", rfe.ranking_)

# Embedded – Random Forest importances
rf = RandomForestClassifier().fit(X, y)
importances = pd.Series(rf.feature_importances_, index=X.columns)
top_features = importances.nlargest(10).index.tolist()

# Embedded – LASSO drives coefficients to zero
lasso = Lasso(alpha=0.01).fit(X_scaled, y)
selected = X.columns[lasso.coef_ != 0]
```

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Outlier Detection and Removal

What are Outliers?

Outliers are data points that significantly deviate from the rest of the dataset. They may arise from genuine rare events, measurement errors, or data-entry mistakes. They can disproportionately skew statistical summaries (mean, variance) and degrade model performance.

Sources of Outliers

- Data entry errors (typed "3000" instead of "30")
- Measurement / instrument failure
- Genuine rare events (fraud, anomaly)
- Sampling errors (wrong group included)
- Natural variability (e.g. height of 7 ft)

Impact on ML Models

- Skews mean and standard deviation
- Linear regression coefficients pulled toward outliers
- K-Means cluster centroids shift
- Reduces overall prediction accuracy
- Misleading visualizations (huge axis range)

Outliers ≠ Errors — When NOT to Remove

In fraud detection, the outliers ARE the signal. In rare-disease prediction, the outliers ARE the target. In anomaly detection, the outliers ARE what we're trying to find. ***Before removing any outlier, ask: is this a measurement error, or is it a genuine rare event with information?*** When in doubt, build TWO models — one with, one without the outliers — and compare.

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Outlier Detection Methods

1. Z-Score Method

Compute z = (x − μ) / σ. Flag points with |z| > 3 as outliers (assumes normal distribution).

```
z = (df - df.mean()) / df.std()
outliers = df[(z.abs() > 3).any(axis=1)]
```

2. IQR Method (Box Plot)

Compute Q1 (25%), Q3 (75%), IQR = Q3 − Q1. Flag points outside [Q1 − 1.5·IQR, Q3 + 1.5·IQR].

```
Q1, Q3 = df.quantile([0.25, 0.75])
IQR = Q3 - Q1
outliers = df[(df < Q1 - 1.5*IQR) | (df > Q3 +
1.5*IQR)]
```

3. Isolation Forest

Tree-based unsupervised algorithm. Outliers are easier to isolate (need fewer splits). Best for high-dimensional data.

```
from sklearn.ensemble import IsolationForest
clf = IsolationForest(contamination=0.05)
preds = clf.fit_predict(X)  # -1 = outlier
```

4. DBSCAN Clustering

Density-based clustering. Points not assigned to any cluster (label = -1) are marked as outliers.

```
from sklearn.cluster import DBSCAN
db = DBSCAN(eps=0.5, min_samples=5).fit(X)
outliers = X[db.labels_ == -1]
```

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Outlier Treatment Options

Once detected, you have four main strategies:

1. Remove (Delete)

Drop the outlier rows. Safe only if outliers are clearly errors and dataset is large.

```
df = df[(z.abs() < 3).all(axis=1)]
```

2. Cap / Winsorize

Replace extreme values with the 5th and 95th percentile (or 1st/99th). Preserves data points.

```
df['col'] = df['col'].clip(lower=p5, upper=p95)
```

3. Transform (Log / Sqrt)

Apply a transformation that shrinks the upper tail. Often log or Box-Cox for skewed data.

```
df['col_log'] = np.log1p(df['col'])
```

4. Robust Models

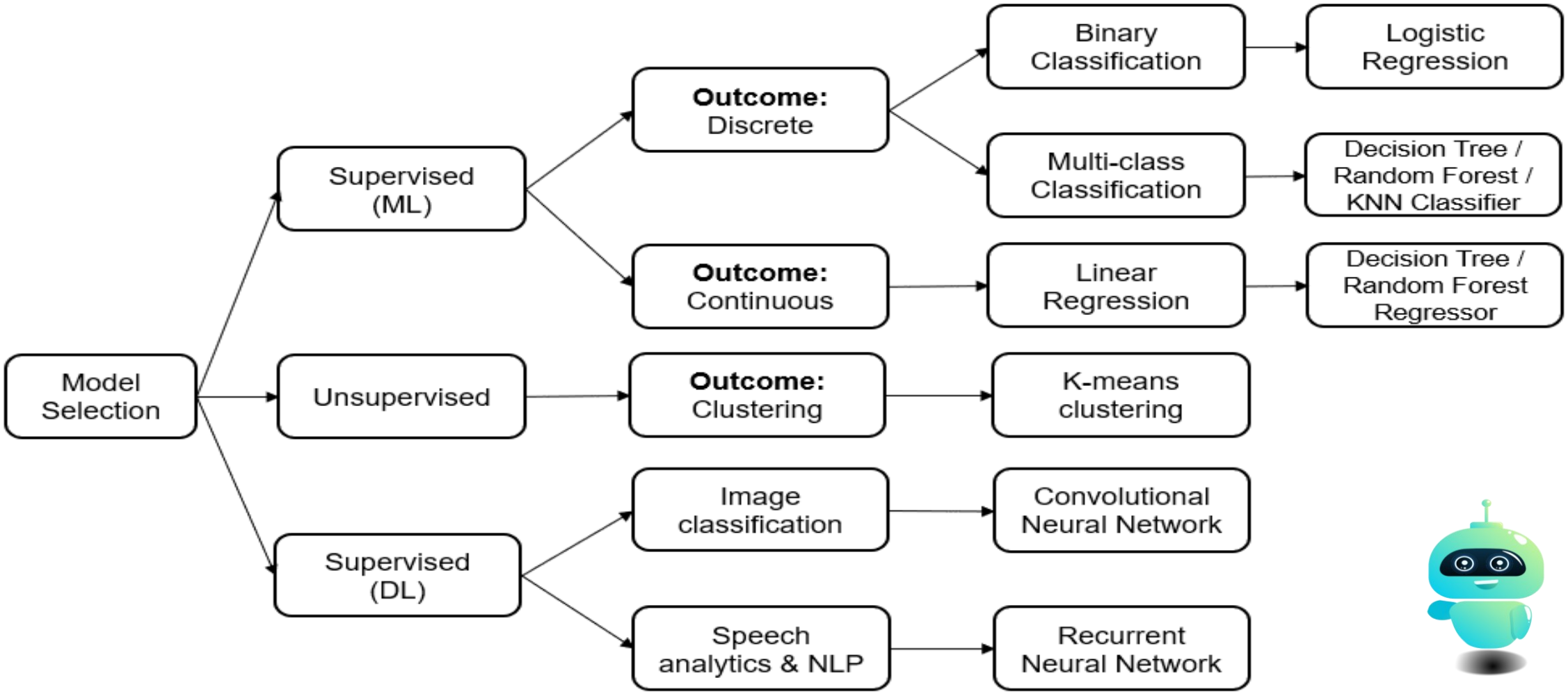
Switch to robust algorithms — RANSAC regression, Huber regression, tree-based models.

```
from sklearn.linear_model import HuberRegressor
model = HuberRegressor().fit(X, y)
```

Pick the strategy that preserves SIGNAL while reducing NOISE — not all outliers should disappear.

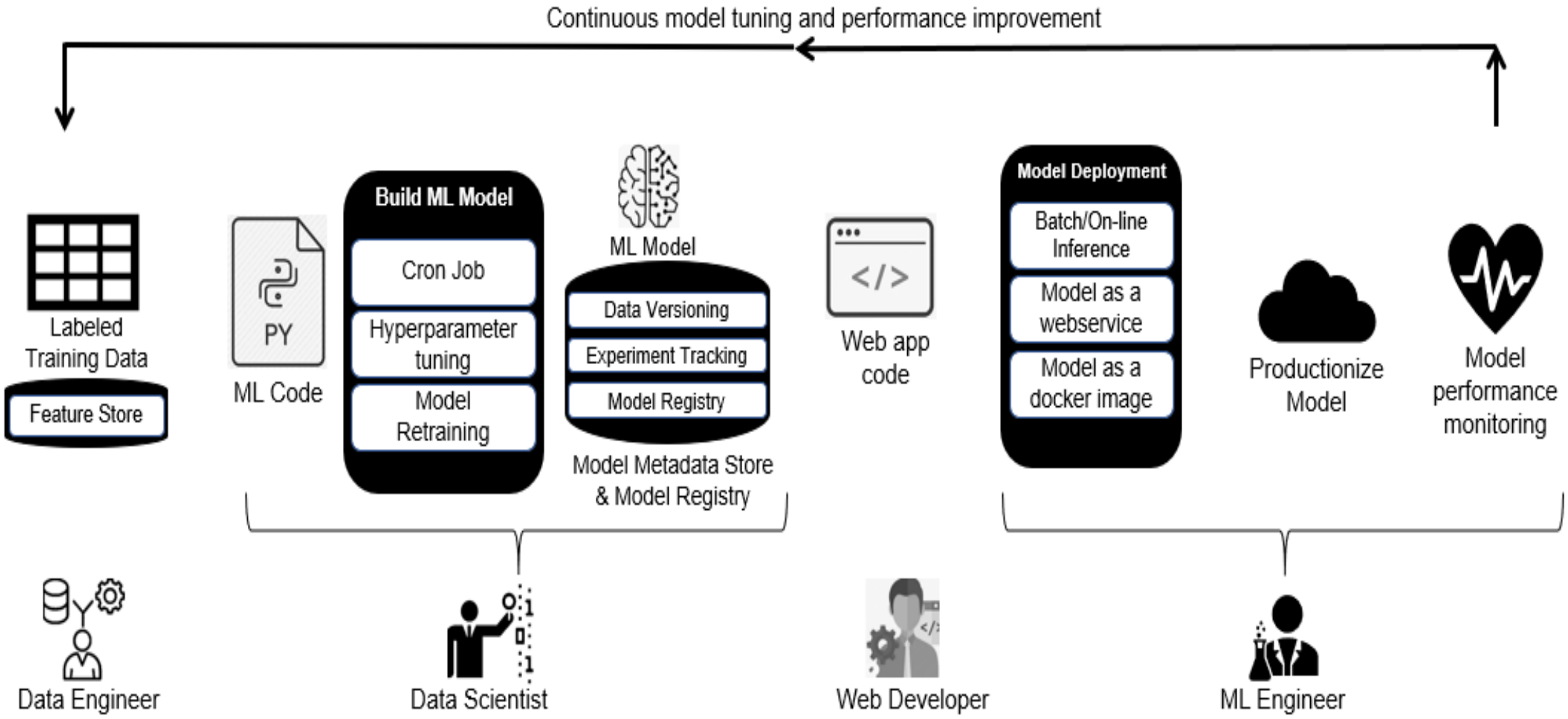
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Machine Learning Model Building



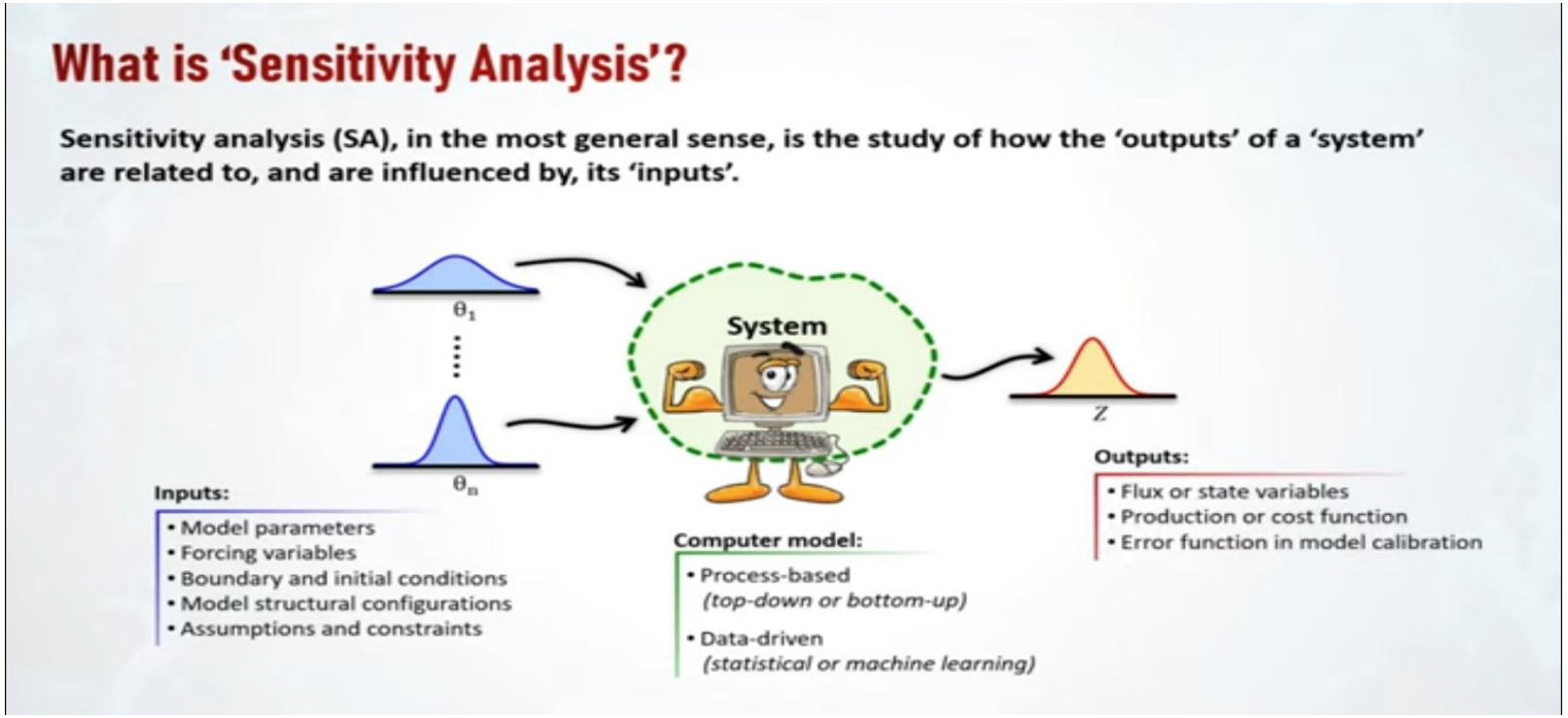
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Machine Learning Model Building



Sensitivity analysis

Sensitivity analysis is a technique used to **understand how changes in the input variables (features) affect the output (predictions) of a model**.
It checks **which features are most influential** in determining the prediction.
It tells us **how robust** our model is when inputs change slightly.



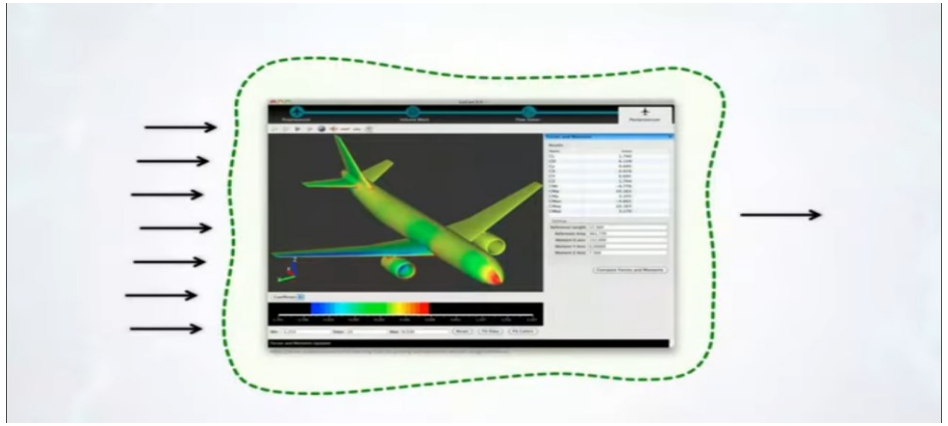
It's very useful for:

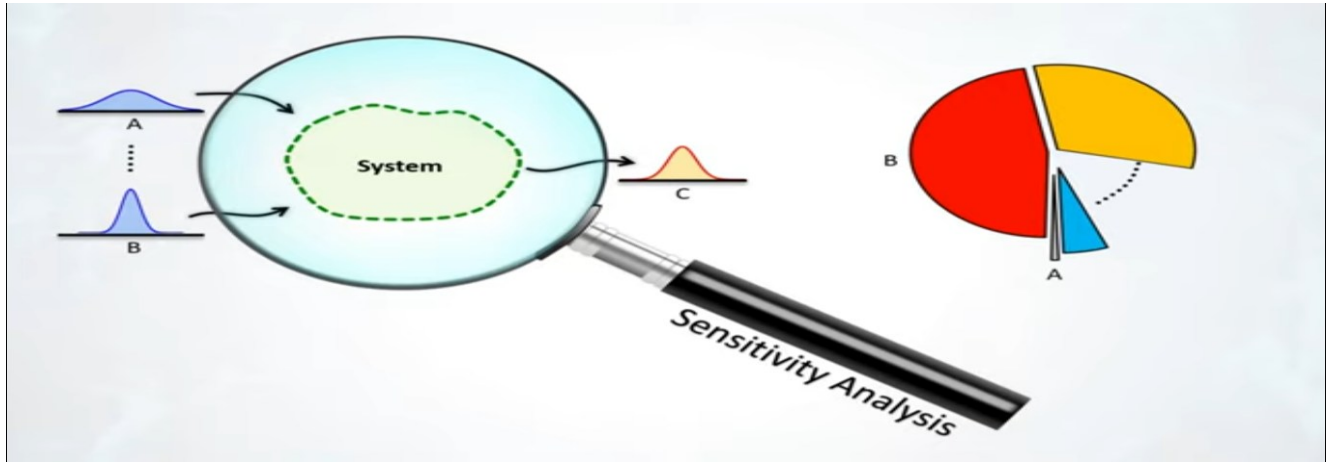
Model interpretation (feature importance).
Debugging models (seeing if the model is too sensitive to noise).
Decision-making (understanding which inputs matter the most).

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Sensitivity analysis



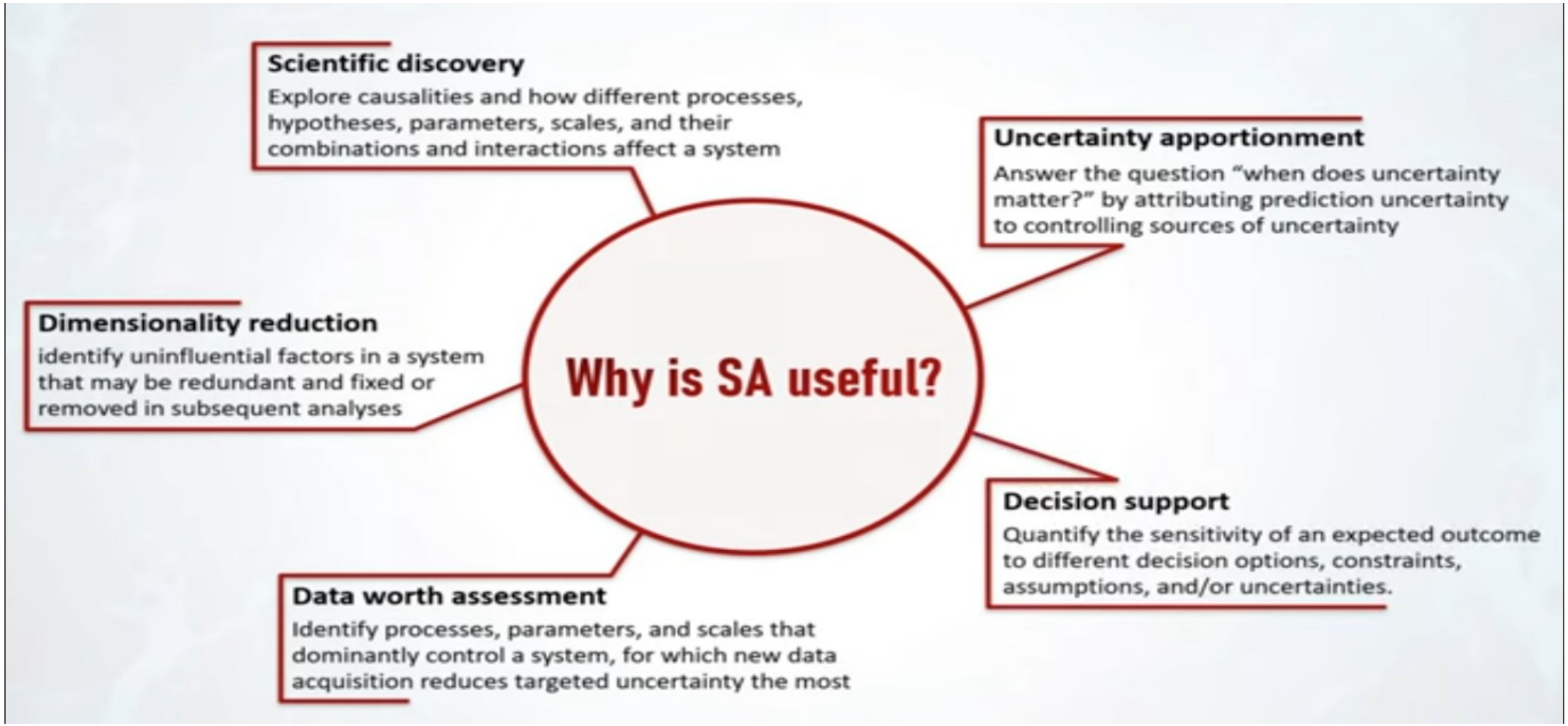




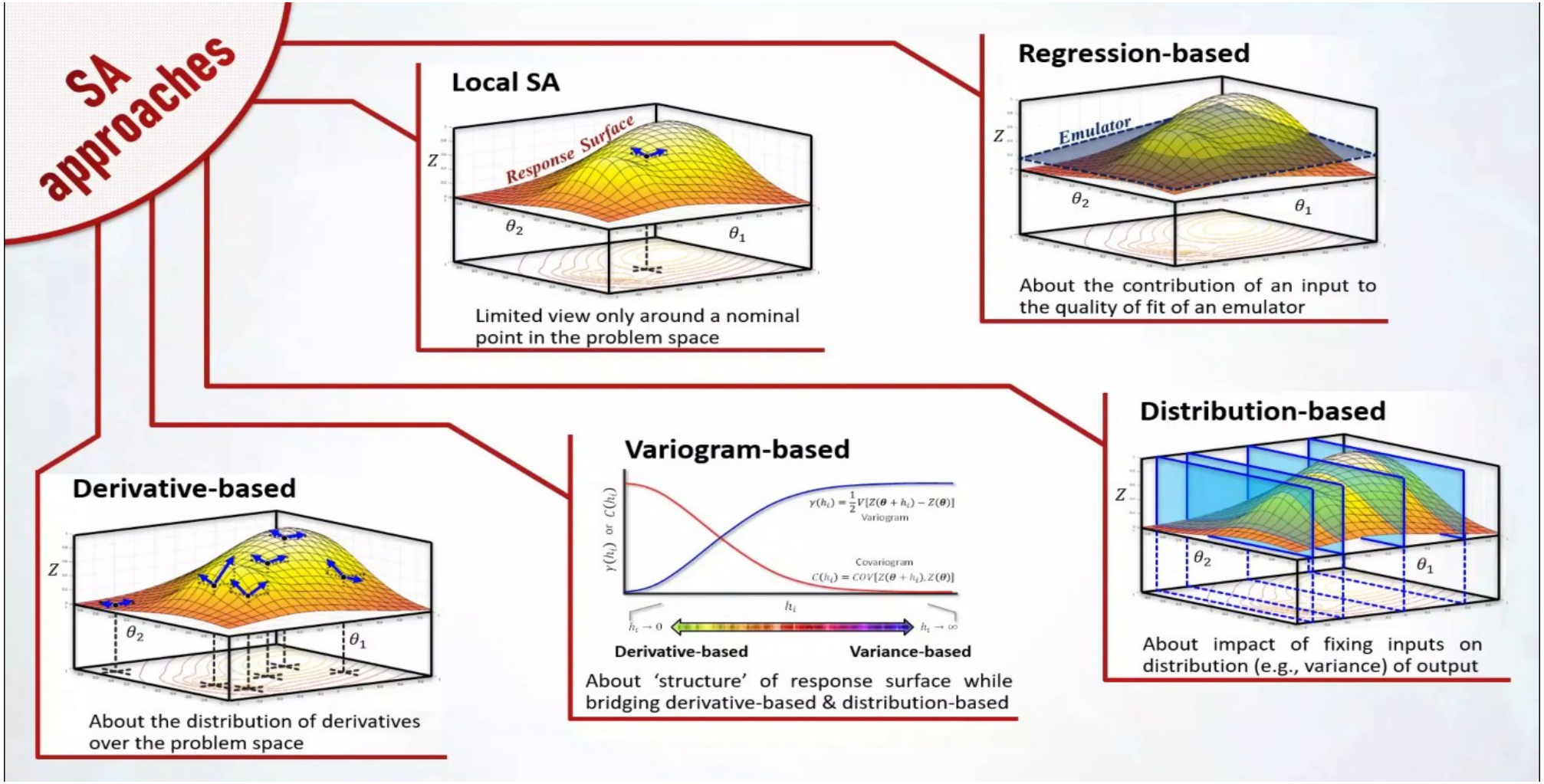


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Sensitivity analysis



Sensitivity analysis



Sensitivity analysis

Types of Sensitivity Analysis in ML

1.Local Sensitivity Analysis (One-at-a-Time - OAT)

1. Only **one input feature is varied** at a time, keeping others fixed.
2. Checks how small changes in one feature affect the prediction.
3. Example: In a house price model, increase "area" by 10% while keeping other features constant, and see how the price prediction changes.
4. **Advantage:** Simple and easy to interpret.
5. **Limitation:** Does not capture interactions between features.

2. Global Sensitivity Analysis

All input features are varied **together** (across their ranges).

Measures the **overall influence** of each feature on the output.

Example: Vary all features (area, location, number of rooms) simultaneously to see the combined effect on house price.

Methods:

Variance-based methods (Sobol indices): Check how much of the output variance is explained by each input.

Monte Carlo sampling: Randomly sample inputs and analyze output changes.

Advantage: Captures interactions and gives a complete picture.

Limitation: Computationally expensive.

Sensitivity analysis

3. Deterministic vs. Probabilistic Sensitivity Analysis

Deterministic: Change inputs by fixed amounts (e.g., +10%, -10%).

Probabilistic: Treat inputs as random variables with distributions, then check output sensitivity (useful in uncertain environments).

4. Local Derivative-based Sensitivity

Uses **partial derivatives** of the model output with respect to inputs.

Tells us how much the prediction will change if a feature changes slightly.

Often used in **deep learning** with gradient-based methods.



Sensitivity analysis

5. Scenario-based Sensitivity

Instead of mathematical variation, you test **realistic scenarios**.

Example: "What happens if interest rates go up by 2%?" in a loan default prediction model.

6. Variogram-based Sensitivity Analysis

A **variogram** measures the **variability (variance) of output differences** as input changes.

Idea: Instead of looking at regression coefficients, it checks **how much the output changes when inputs are perturbed**(altered a little).

Often used in **geo-statistics** and **environmental ML models** (e.g., rainfall prediction, spatial modelling).

Example:

If a small change in rainfall input causes large differences in river flow prediction, the variogram will show **high sensitivity**.

Under Fitting and Over Fitting

Overfitting and Underfitting are the two main problems that occur in machine learning and degrade the performance of the machine learning models.

generalization defines the ability of an ML model to provide a suitable output by adapting the given set of unknown input. underfitting and overfitting are the two terms that need to be checked for the performance of the model and whether the model is generalizing well or not.

- **Signal:** It refers to the true underlying pattern of the data that helps the machine learning model to learn from the data.
- **Noise:** Noise is unnecessary and irrelevant data that reduces the performance of the model.
- **Bias:** Bias is a prediction error that is introduced in the model due to oversimplifying the machine learning algorithms. Or it is the **difference between the predicted values and the actual values**.
- **Variance:** If the machine learning model performs well with the training dataset, but does not perform well with the test dataset, then variance occurs.

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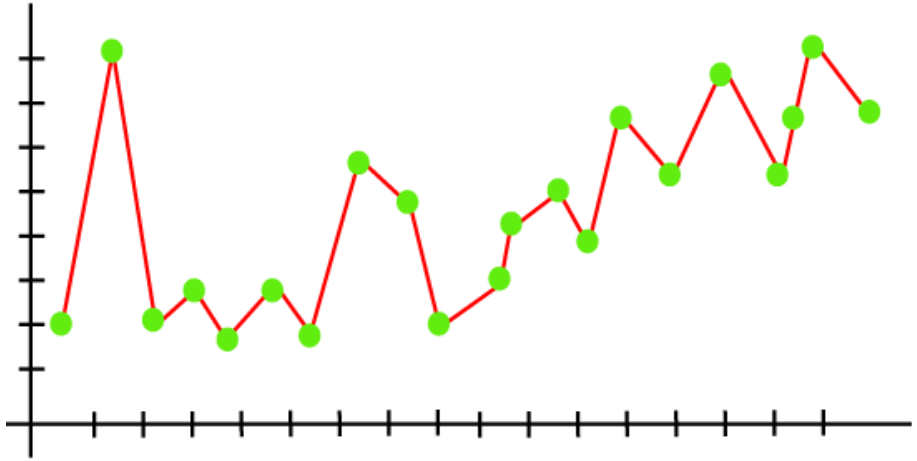
Under Fitting and Over Fitting

Overfitting

Overfitting occurs when our machine learning model tries to cover all the data points, or more than the required data points present in the given dataset.

How to avoid the Overfitting in Model

Cross-Validation
Training with more data
Removing features
Early stopping the training
Regularization
Ensemble Learning (multiple learning algos to obtain better predictive performance)
-Bagging, Boosting, Stacking



The chances of occurrence of overfitting increase as much we provide training to our model. It means the more we train our model, the more chances of occurring the overfitted model.

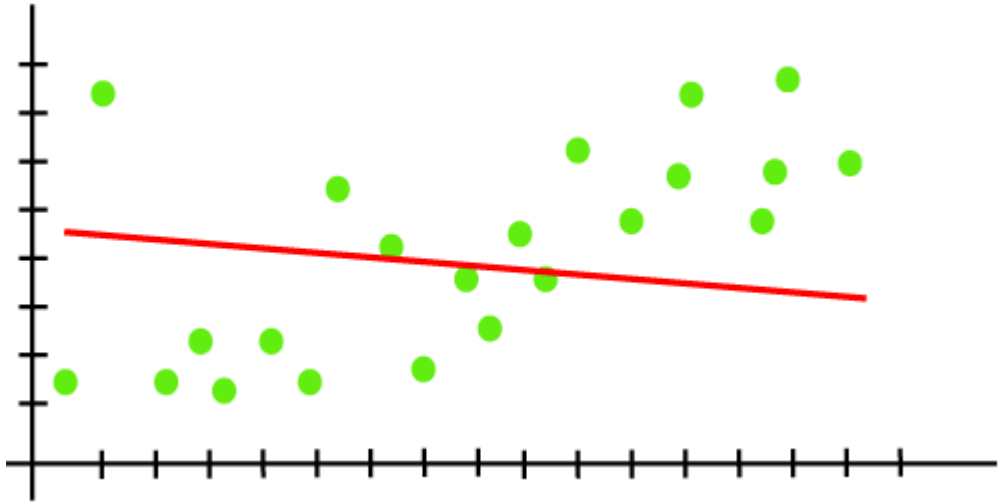
Under Fitting and Over Fitting

Underfitting

Underfitting occurs when our machine learning model is not able to capture the underlying trend of the data.

How to avoid underfitting:

By increasing the training time of the model.
By increasing the number of features.



An underfitted model has high bias and low variance.

Goodness of Fit

The "Goodness of fit" term is taken from the statistics, and the goal of the machine learning models to achieve the goodness of fit. In statistics modeling, it defines how closely the result or predicted values match the true values of the dataset.

K-Fold Cross Validation

K-Fold Cross-Validation

- **The dataset is divided into K equally sized subsets or folds.**
- **The model is trained on K-1 folds and tested on the remaining fold.**
- **This process is repeated K times, with each fold serving as the test set once.**

[16, 22, 31, 47, 59, 63, 77, 81, 90, 100, 111, 124]

K=4

Divide the dataset into 4 equal parts, K1, K2, K3, K4.

I_1 = K1-K2-K3 training / K4 testing	80%
I_2 = K1-K2-K4 training / K3 testing	70%
I_3 = K1-K3-K4 training / K2 testing	75%
I_4 = K2-K3-K4 training / K1 testing	90%

Average performance
(80+70+75+90)/4
=



Stratified K-Fold Cross Validation

Stratified K-Fold Cross-Validation

Similar to K-Fold Cross-Validation, but it ensures that each fold has the same proportion of data for each class label (important for imbalanced datasets).

[10, 21, 33, 14, 35, 66, 87, 89, 29, 110, 11, 112, 23, 43, 28, 190]

[A, A, A, A, A, A, A, A, B, B, B, B, B, B, B, B]

K=4 dataset contains 50% of A and 50% B class

Divide the dataset into 4 parts containing both A and B (50%),

K1(10,21,29,110), K2(33,14,11,112), K3(35,66,23,43), K4(87,89,28,190).

I_1 = K1-K2-K3 training / K4 testing

I_2 = K1-K2-K4 training / K3 testing

I_3 = K1-K3-K4 training / K2 testing

I_4 = K2-K3-K4 training / K1 testing



Leave-One-Out Cross Validation (LOOCV)

Leave-One-Out Cross-Validation (LOOCV)

In this method, each individual data point is used once as a test set while all other data points are used for training.

This means if you have n data points, the process is repeated n times.

[10, 21, 33, 14, 35, 66, 87, 89, 29, 110, 11, 112, 23, 43, 28, 190]

K=1

Divide the dataset into 16 parts,

I_1 = K2-K16 training / K1 testing

I_2 = K1,K3-K16 training / K2 testing

I_3 = K1-K2,K4-K16 training / K3 testing

So on....



Bias and Variance in Machine Learning

What is Bias?

Bias is simply defined as the inability of the model because of that there is some difference or error occurring between the model's predicted value and the actual value. These differences between actual or expected values and the predicted values are known as error or bias error or error due to bias. Bias is a systematic error that occurs due to wrong assumptions in the machine learning process.

Let Y be the true value of a parameter and let Y^ be an estimator of Y based on a sample of data. Then, the bias of the estimator Y^ is given by:

$$\text{Bias}(\hat{Y}) = E(\hat{Y}) - Y$$

- **Low Bias:** Low bias value means fewer assumptions are taken to build the target function. In this case, the model will closely match the training dataset.
- **High Bias:** High bias value means more assumptions are taken to build the target function. In this case, the model will not match the training dataset closely.

Bias and Variance in Machine Learning

What is Variance?

Variance is the measure of spread in data from its mean position. In machine learning variance is the amount by which the performance of a predictive model changes when it is trained on different subsets of the training data. More specifically, variance is the variability of the model that how much it is sensitive to another subset of the training dataset. i.e. how much it can adjust on the new subset of the training dataset.

Let Y be the actual values of the target variable, and Y^ be the predicted values of the target variable. Then the variance of a model can be measured as the expected value of the square of the difference between predicted values and the expected value of the predicted values.

$$\text{Variance}=E[(\hat{Y}-E[\hat{Y}])2]$$

- **Low variance:** Low variance means that the model is less sensitive to changes in the training data and can produce consistent estimates of the target function with different subsets of data from the same distribution. This is the case of underfitting when the model fails to generalize on both training and test data.
- **High variance:** High variance means that the model is very sensitive to changes in the training data and can result in significant changes in the estimate of the target function when trained on different subsets of data from the same distribution. This is the case of overfitting when the model performs well on the training data but poorly on new, unseen test data. It fits the training data too closely that it fails on the new training dataset.

Bias-Variance Tradeoff

The Tradeoff

Reducing bias usually INCREASES variance and vice versa. The art of ML modeling is finding the SWEET SPOT — enough complexity to capture true patterns but not so much that noise is memorized.

Dartboard Analogy

Low Bias, Low Variance

Predictions ON-TARGET and TIGHTLY clustered.

Ideal — what we strive for.

Low Bias, High Variance

On-target on AVERAGE but SCATTERED widely.

Overfitting — model is too sensitive.

High Bias, Low Variance

Tightly clustered but FAR from the target.

Underfitting — model is consistently wrong.

High Bias, High Variance

Far from target AND scattered widely.

Worst case — model is wrong and unstable.

Sweet spot: a model complex enough to capture the signal but not the noise.

Problem 1: Feature Types, Missing Data, Categorical Features and Feature Scaling

Problem Statement

Consider the following dataset of students applying for internships.

ID	Age	Gender	CGPA	Income (Rs.)	Selected
1	20	Male	8.2	25000	Yes
2	21	Female	7.8	30000	No
3	22	Male	?	28000	Yes
4	20	Female	9.1	35000	Yes
5	?	Female	8.0	27000	No

Perform the following:

1. Identify feature types.
2. Handle missing values.
3. Encode categorical variables.
4. Perform Min-Max Scaling on Age.

Problem 2: Feature Selection using Chi-Square Test

Problem Statement

A dataset contains two categorical features.

Feature	Chi-Square Score
Attendance	12.5
Sports Participation	2.1

Critical Chi-square value at $\alpha = 0.05$ is

$$3.84$$

Determine which feature should be selected.

Problem 3: Outlier Detection and Removal using Z-Score

Problem Statement

A teacher collected students' marks.

$$40,\ 42,\ 39,\ 41,\ 43,\ 40,\ 150$$

Determine whether 150 is an outlier using the Z-score method.
Use

$$|Z| > 3$$

to identify an outlier.

Mathematical Decomposition

For a model $\hat{f}(x)$ trying to predict $y = f(x) + \varepsilon$:

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2[\hat{f}(x)] + \text{Var}[\hat{f}(x)] + \sigma^2\varepsilon$$

Bias²

$$(E[\hat{f}(x)] - f(x))^2$$

How far our average prediction is from the truth. A model that's systematically wrong.

Variance

$$E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

How much our prediction varies if we re-train on different samples. Model instability.

Irreducible Noise ($\sigma^2\varepsilon$)

$$\text{Var}(\varepsilon)$$

Inherent randomness in the data — no model can eliminate this. Sets a lower bound on error.

Finding the Sweet Spot

As model complexity increases, BIAS² decreases monotonically, but VARIANCE increases. Total error is U-shaped — the optimum is where bias and variance are balanced. Validation error guides you there.

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Activation Functions

Why Activation Functions?

Neural networks compute weighted sums of inputs: $z = \Sigma\, w \cdot x + b$. Without a NON-LINEAR activation, a stack of layers reduces to a single linear function — no matter how deep. Activation functions inject non-linearity, enabling networks to learn complex patterns.

Four Common Activation Families

Sigmoid

S-curve, output in (0, 1)
Binary output

Tanh

S-curve, output in (-1, 1)
Zero-centered, RNN hidden

ReLU

max(0, x), output in [0, ∞)
Fast, default for CNN/MLP

Softmax

Multi-class probability
Final layer for classification

Quick Proof — Why Non-Linearity Matters

Consider 2 linear layers without activation:

$$\text{Layer 2}\,(\,\text{Layer 1}\,(x)\,) = W_2\,(W_1\,x + b_1) + b_2 = (W_2\,W_1)\,x + (W_2\,b_1 + b_2) = Wx + b$$

Two linear layers behave EXACTLY like one. Without activations, depth gives nothing!

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Sigmoid & Tanh

Sigmoid (Logistic)

$$\sigma(x) = 1 / (1 + e^{(-x)})$$

Range: `(0, 1)`

Pros:

- Smooth, differentiable everywhere
- Output interpretable as probability
- Maps any real input to a bounded range

Cons:

- Vanishing gradient for |x| > 3
- Output not zero-centered
- exp() is computationally heavy

Best for:

binary classification output layer.

Tanh (Hyperbolic Tangent)

$$\tanh(x) = (e^{x} - e^{-x}) / (e^{x} + e^{-x})$$

Range: `(-1, 1)`

Pros:

- Zero-centered output
- Faster convergence than Sigmoid
- Stronger gradient near 0

Cons:

- Still saturates for large |x|
- Vanishing gradient on the tails
- exp() is still expensive

Best for:

RNN / LSTM hidden states.

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ReLU and its Variants

ReLU (Rectified Linear Unit) — Krizhevsky et al., 2012

$$\text{ReLU}(x) = \max(0, x)$$

Pros: Computationally cheap. No vanishing gradient for x > 0. Sparse activations. Default choice for modern deep networks.
Cons: "Dying ReLU" — once a neuron outputs 0, gradient is 0 and it stops learning. Not zero-centered.

Modern Variants — Fixing ReLU's Issues

Leaky ReLU

$$\max(0.01 \cdot x, x)$$

Small slope for negatives prevents dying neurons.

Parametric ReLU

$$\max(\alpha \cdot x, x)$$

α is LEARNED from data — adaptive negative slope.

ELU

$$x \text{ if } x > 0 \text{ else } \alpha(e^x - 1)$$

Smooth negative side, robust to noise, mean closer to 0.

GELU

$$x \cdot \Phi(x)$$

Probabilistic gating. Used in BERT, GPT, all modern Transformers.

Modern Default: ReLU for CNNs and MLPs. GELU for Transformers. Leaky ReLU when dying ReLU becomes a problem.

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Softmax & Comparison

Softmax

$$\text{softmax}(z_i) = e^{(z_i)} / \Sigma_j e^{(z_j)}$$

Converts K logits into K probabilities that SUM TO 1. Used at the OUTPUT layer of multi-class classifiers.

Example: `[2.0, 1.0, 0.1]` → `[0.66, 0.24, 0.10]` → `predicted class is 0.`

Comparison of Activation Functions

Function	Range	Pros	Cons	Use For
Sigmoid	(0, 1)	Probabilistic	Vanishing gradient	Binary output
Tanh	(-1, 1)	Zero-centered	Still saturates	RNN hidden states
ReLU	[0, ∞)	Fast, sparse	Dying ReLU	CNN, MLP hidden
Leaky ReLU	(-∞, ∞)	Fixes dying ReLU	Slightly slower	Robust alternative
GELU	(-∞, ∞)	Smooth, probabilistic	Heavier compute	BERT, GPT, all Transformers
Softmax	(0, 1), sum=1	Multi-class probabilities	Output layer only	Multi-class output

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Data Science vs Machine Learning

Two related-but-distinct disciplines that are often confused:

Data Science

Definition

An interdisciplinary field that extracts INSIGHTS from structured and unstructured data using statistics, programming, and domain expertise.

Goal

Understand the data — descriptive, exploratory, and prescriptive analytics.

Typical Activities

- Data cleaning, EDA, dashboards
- Statistical testing & inference
- Visualization & storytelling
- Business intelligence reporting
- Building ML models (subset)

Output

Reports, dashboards, recommendations.

Machine Learning

Definition

A SUBFIELD of AI focused on algorithms that LEARN PATTERNS from data and improve performance with experience.

Goal

Build predictive models that generalize to unseen data.

Typical Activities

- Feature engineering
- Training algorithms (SVM, NN, RF)
- Hyperparameter tuning
- Model evaluation and deployment
- Monitoring drift in production

Output

Trained models serving predictions.

How They Overlap & Differ

The Relationship

ML is one of many TOOLS in the data scientist's toolkit. Conversely, ML engineers use Data Science skills (EDA, statistics) constantly. Think of Data Science as a SUPERSET that includes ML, but also includes BI, dashboards, statistical inference, and storytelling.

Side-by-Side Comparison

Aspect	Data Science	Machine Learning
Scope	Broader — entire data lifecycle	Narrower — predictive modelling
Primary Goal	Insight & understanding	Prediction & automation
Skills Focus	Statistics, SQL, visualization, business	Algorithms, optimization, deep learning
Outputs	Reports, dashboards, recommendations	Trained models, prediction APIs
Typical Tools	Tableau, Power BI, SQL, Pandas, R	scikit-learn, TensorFlow, PyTorch
Background	Stats + business analytics	CS + applied math
Team Size	Often works solo or in small teams	Engineering team with MLOps support

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Assignment 1

1. Define Machine Learning. Discuss with examples why machine learning is important.
2. Discuss with examples some useful applications of machine learning.
3. Explain how some areas/disciplines that influenced the machine learning.
4. What do you mean by a well–posed learning problem? Explain the important features that are required to well–define a learning problem.
5. Define learning program for a given problem. Describe the following problems with respect to Tasks, Performance and Experience:
   a. Checkers Learning Problems
   b. Handwritten Recognition Problem
   c. Robot Driving Learning Problem
6. Describe in detail all the steps involved in designing a learning system.
7. Discuss the perspective and issues in machine learning.

- NOTE: All Students have to write this assignment in separate note book.

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Assignment 2

1. What is Artificial Neural Network?
2. What are the type of problems in which Artificial Neural Network can be applied?
3. What is difference between BNN and ANN ?
4. Explain the k-Means Algorithm with an example.
5. What is Reinforcement Learning? Explain with the help of example.
6. Explain Bayesian networks with the help of Examples.
7. Explain Support Vector Machine in details. Describe their importance.
8. Explain Genetic Algorithms in details.

Daily Quiz

1. What is the main task of a problem-solving agent?
a) Solve the given problem and reach to goal
b) To find out which sequence of action will get it to the goal state
c) All of the mentioned
d) None of the mentioned

2. What is state space?
a) The whole problem
b) Your Definition to a problem
c) Problem you design
d) Representing your problem with variable and parameter

Daily Quiz

3. The problem-solving agent with several immediate options of unknown value can decide what to do by just examining different possible sequences of actions that lead to states of known value, and then choosing the best sequence. This process of looking for such a sequence is called Search.

a) True

b) False

4. A search algorithm takes _____ as an input and returns _____ as an output.

a) Input, output

b) Problem, solution

c) Solution, problem

d) Parameters, sequence of actions

Daily Quiz

5. A problem in a search space is defined by one of these state.
a) Initial state
b) Last state
c) Intermediate state
d) All of the mentioned

6. The Set of actions for a problem in a state space is formulated by a _____
a) Intermediate states
b) Initial state
c) Successor function, which takes current action and returns next immediate state
d) None of the mentioned

MCQ

1. Web Crawler is a/an _____
a) Intelligent goal-based agent
b) Problem-solving agent
c) Simple reflex agent
d) Model based agent

2. What is the major component/components for measuring the performance of problem solving?
a) Completeness
b) Optimality
c) Time and Space complexity
d) All of the mentioned

MCQ

3. A production rule consists of _____
a) A set of Rule
b) A sequence of steps
c) Set of Rule & sequence of steps
d) Arbitrary representation to problem

4. Which search method takes less memory?
a) Depth-First Search
b) Breadth-First search
c) Linear Search
d) Optimal search

MCQ

5. Which is the best way to go for Game playing problem?
a) Linear approach
b) Heuristic approach (Some knowledge is stored)
c) Random approach
d) An Optimal approach

6. Which of the following applications include in the Strategic Computing Program?
a) battle management
b) autonomous systems
c) pilot's associate
d) all of the mentioned

MCQ

7. How many things are concerned in the design of a learning element?
a) 1
b) 2
c) 3
d) 4

8. What is used in determining the nature of the learning problem?
a) Environment
b) Feedback
c) Problem
d) All of the mentioned

MCQ

9. How many types are available in machine learning?
a) 1
b) 2
c) 3
d) 4

10. Which is used for utility functions in game playing algorithm?
a) Linear polynomial
b) Weighted polynomial
c) Polynomial
d) Linear weighted polynomial

Glossary Questions

1. A solution to a problem is a ……from the initial state to a goal state. Solution quality is measured by the path cost function, and an optimal solution has the highest path cost among all solutions.
a) Path
b) Road

2. The process of removing detail from a given state representation is called _____
a) Extraction
b) Abstraction
c) Information Retrieval
d) Mining of data

Glossary Questions

3. A problem solving approach works well for _____
a) 8-Puzzle problem
b) 8-queen problem
c) Finding a optimal path from a given source to a destination
d) Mars Hover (Robot Navigation)

4. The _____ is a touring problem in which each city must be visited exactly once. The aim is to find the shortest tour.
a) Finding shortest path between a source and a destination
b) Travelling Salesman problem
c) Map coloring problem
d) Depth first search traversal on a given map represented as a graph

Weekly Assignment

Assignment 1

- Describe Machine Learning with suitable examples.[CO1]
- Brief the well defined learning problems. [CO1]
- Define the process of Designing a Learning System? [CO1]
- Explain different perspective and Issues in Machine Learning. [CO1]
- Define hypotheses. [CO1]
- Analyze THE CONCEPT LEARNING TASK - General-to-specific ordering of hypotheses? [CO1]
- Illustrate Find-S algorithm. [CO1]
- Describe List-Then-Eliminate algorithm. [CO1]
- Define Candidate elimination algorithm with example. [CO1]
- Briefly explain Inductive bias. [CO1]

Old Question Papers

Note: No old question paper available for this subject. Introduced first time.

I have added expected question for university exam in next slide.

Expected Questions for University Exam

- Illustrate Find-S algorithm. [CO1]
- Describe List-Then-Eliminate algorithm. [CO1]
- Define Candidate elimination algorithm with example. [CO1]
- Briefly explain Inductive bias. [CO1]
- Describe Machine Learning with suitable examples.[CO1]
- Brief the well defined learning problems. [CO1]
- Define the process of Designing a Learning System? [CO1]
- Explain different perspective and Issues in Machine Learning. [CO1]
- Define hypotheses. [CO1]
- Analyze THE CONCEPT LEARNING TASK - General-to-specific ordering of hypotheses? [CO1]

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2. Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.

Recap of Unit

- Concept learning can be seen as a problem of searching through a large predefined space of potential hypotheses.
- The general-to-specific partial ordering of hypotheses provides a useful structure for organizing the search through the hypothesis space.
- The FIND-S algorithm utilizes this general-to-specific ordering, performing a specific-to-general search through the hypothesis space along one branch of the partial ordering, to find the most specific hypothesis consistent with the training examples.

Faculty Video Links, Youtube & NPTEL Video Links and Online Courses Details

Youtube video-

- https://www.youtube.com/watch?v=PDYfCkLY_DE
- https://www.youtube.com/watch?v=ncOirIPHTOw
- https://www.youtube.com/watch?v=cW03t3aZkmE

Thank you